

## INTRODUCING BOND SPREADS, DURATION & CONVEXITY

### Global Credit Derivative and Index Research

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J.P. Morgan Securities plc

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## Bond prices and yields

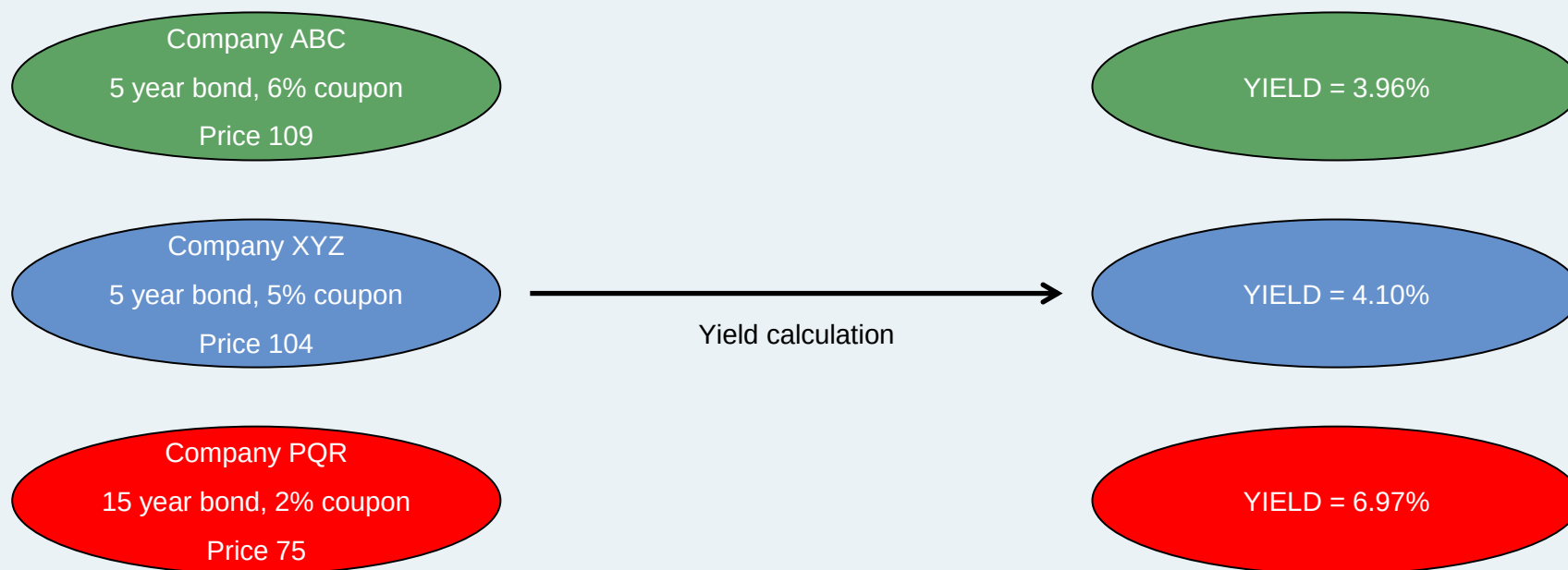
### Bond pricing

- The yield of a bond is the average return on investment from buying a bond.

- The price of a bond can be expressed in terms of its yield:

$$\text{Price} = \frac{C}{1+y} + \frac{C}{(1+y)^2} + \frac{C}{(1+y)^3} + \dots + \frac{100+C}{(1+y)^T}$$

- Higher yield → lower prices.
- Yields allows us to easily compare bonds with different coupons, prices and maturities.

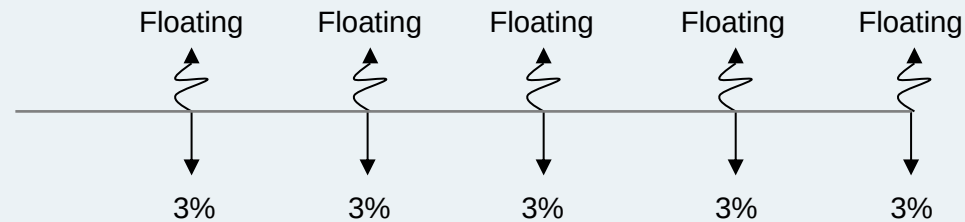


Source: J.P. Morgan

## Interest rate swaps

### Swaps

- Interest rate swaps allow investors to convert floating-rate cash flows to fixed-rate cash flows.
- e.g. for 5y swap we pay fixed rate of 3% every year for five years and receive floating rate payments based upon some reference rate (Euribor/SONIA/SOFR).
  - Future floating rate levels are not yet known – 3% rate is the market's expectation of the average floating rate over the life of the contract.



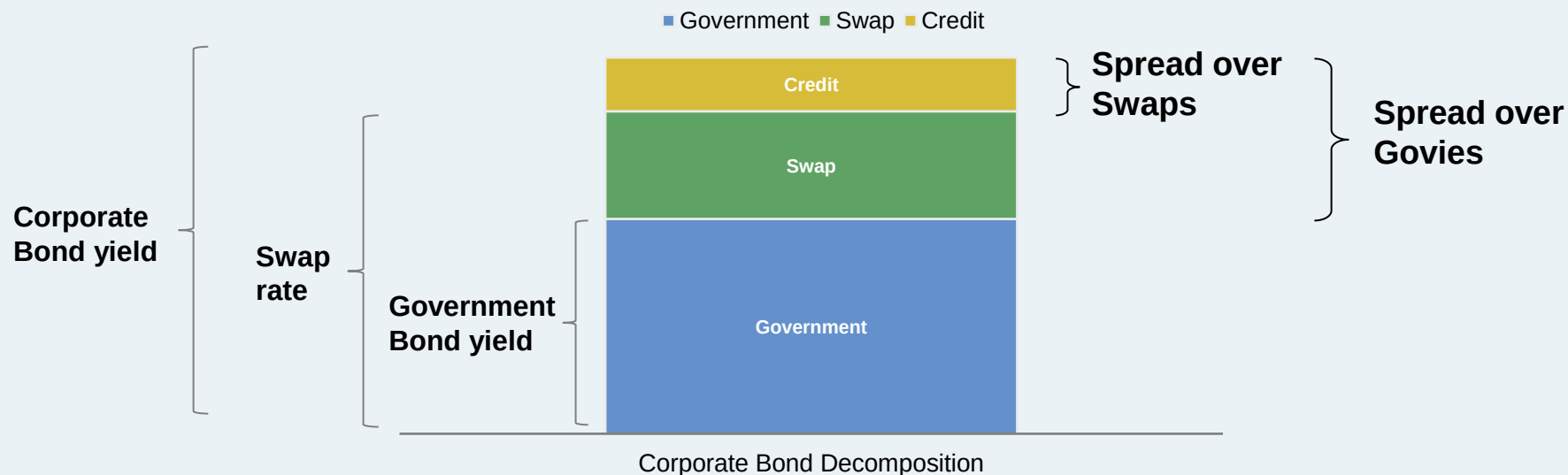
- Reference rates for swaps in some currencies have changed to overnight funding rates, with LIBOR phased out and replaced by new RFRs (SOFR in the US, SONIA in the UK, ESTR in EZ), with exception of Euribor which remained.
- LIBOR has historically been seen as average short-term borrowing rate for AA- rated banks; for financial investors looking to borrow money and invest in higher yielding assets will benchmark themselves to this level.
- Credit instruments are often compared to the swaps rate to ascertain how much relative yield they offer.

Source: J.P. Morgan

## Components of a corporate bond yield

### Corporate bond yields

- The yield is the return for taking on the risk of holding a (corporate) bond
  - For corporate bonds we can break this down into three components
- Each component represents a different risk/reward of the bond
  - The spread is the additional return for taking on additional risk



- Credit spreads can be to government or swaps (i.e. to a given “risk-free” curve”)
  - Swaps are based on interbank lending of generic AA banks (RFRs)
  - Swaps convert floating reference rates to fixed rates

Source: J.P. Morgan

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## Why we only care about spreads

### Yields and Spreads

- Why can't we just use yields?
  - Yield changes can be driven by any one of the three components: government, swap or credit
  - We want to **isolate the credit risk** - i.e. the credit spread
  - Historically yields have been dominated by the risk-free component but since 2008 credit spreads have been a much larger driver.
- For example
  - Company ABC has a bond with a yield of 6%
  - Central bank raises interest rates leading to a 1% rise in government bond yields
  - Company ABC's bond now has a yield of 7%
  - Why has the bond price declined?
    - Company credit quality is unchanged
    - Yield change is due to government yield change
- **Yield changes alone do not tell us about credit quality**

Source: J.P. Morgan

## Credit and risk-free rates

### Risk-free rates

- Credit spreads are always quoted with respect to a risk-free benchmark.
- “Risk-free” can be quite subjective and depends on the investor. Typically there are two broad types of risk-free rates:
  - **Government bonds:** Mostly used by clients who are given cash to invest and do not have to borrow funds, e.g. asset managers, pension funds, insurance companies.
  - **Interest rates swaps:** Mostly used by clients who borrow money to fund themselves; LIBOR swaps historically represented borrowing rate for banks and other financial institutions. e.g. Bank desks, hedge funds.
- Neither government bonds nor interest rate swaps are truly risk free; from a credit point of view we are most interested in them for comparison purposes.

Source: J.P. Morgan



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# 1. Spread to Government

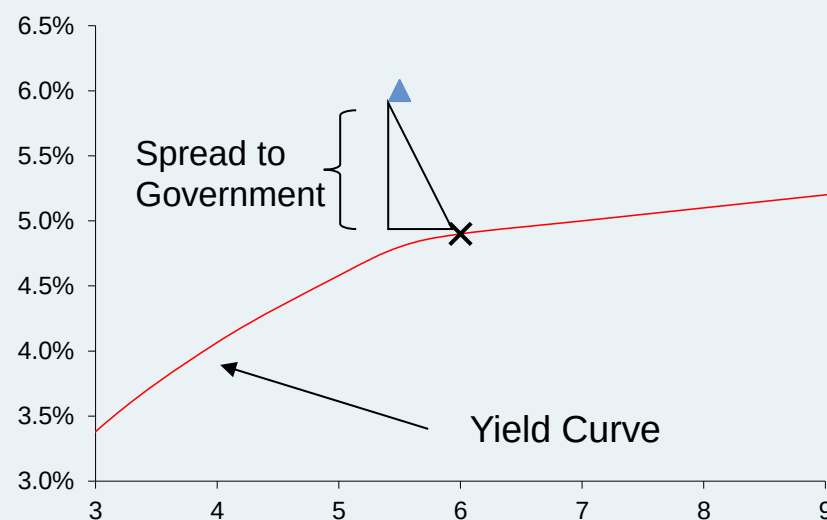
## Spread to Govt

- Spread to government is the difference in yield between a corporate bond and a government bond of a similar maturity.

- Example: Verizon 5.5y \$ bond  
US Treasury 6y \$ bond

Yield: 6% (600bp)  
Yield: 4.9% (490bp)  
**Spread to govt: 110bp**

- Advantages:
  - Easy to calculate and quote
  - Good benchmark for real money investors
- Disadvantages:
  - Maturity mismatch
  - Doesn't take structure of payments into account.



Source: J.P. Morgan

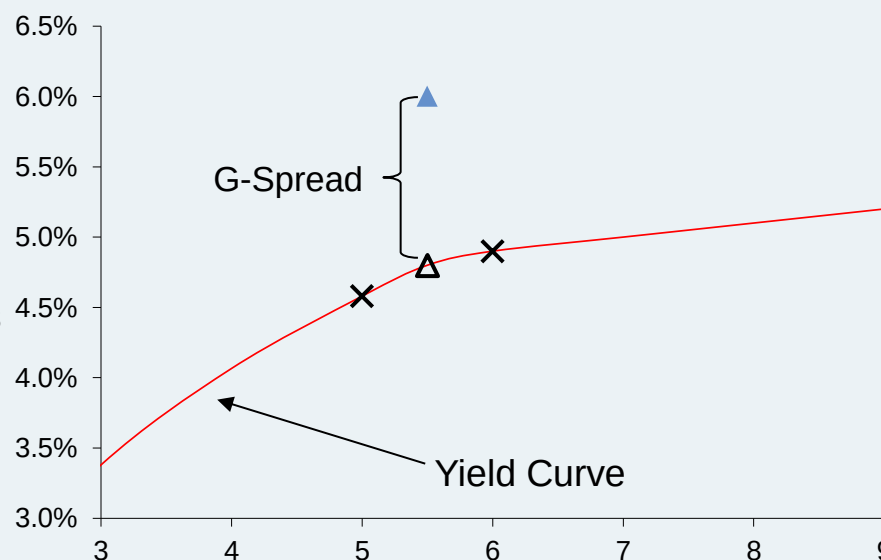
## 2. Interpolated Government Spread

### Interpolated Spread to Govt (G-Spread)

- Instead of calculating the difference between the corp bond yield and the yield of the nearest govt bond, we interpolate along the govt yield curve to calculate the yield that a govt bond would trade with if it had the exact same maturity as the corporate bond.

|                                           |               |             |                |
|-------------------------------------------|---------------|-------------|----------------|
| ■ Example: Verizon 5.5y bond              | Yield:        | 6%          | (600bp)        |
| US Treasury 5y bond                       | Yield:        | 4.6%        | (460bp)        |
| US Treasury 6y bond                       | Yield:        | 4.9%        | (490bp)        |
| <i>Interpolated US 5.5y bond</i>          | <i>Yield:</i> | <i>4.8%</i> | <i>(480bp)</i> |
| <b>Interpolated spread to govt: 120bp</b> |               |             |                |

- Advantages:
  - No maturity mismatch
- Disadvantages:
  - Doesn't take structure of payments into account.
  - Interpolation method might not be obvious



Source: J.P. Morgan

### 3. Z-Spreads

#### Z-Spreads

- We can calculate “risk-free” present value for a bond using the swaps implied zero curve.

$$\text{RiskFreePrice} = \frac{C}{1 + z_1} + \frac{C}{(1 + z_2)^2} + \dots + \frac{100 + C}{(1 + z_T)^T}$$

$z_1, z_2, z_3$  are zero rates implied from swaps curve

- This gives us the price of the bond if it was trading in line with the swaps market.
- Most corporate bonds are actually trading below this price as they are more risky than swaps market: we include an extra discount factor  $s$  so that the price matches that seen in the market.

$$\text{TradedPrice} = \frac{C}{1 + z_1 + s} + \frac{C}{(1 + z_2 + s)^2} + \dots + \frac{100 + C}{(1 + z_T + s)^T}$$

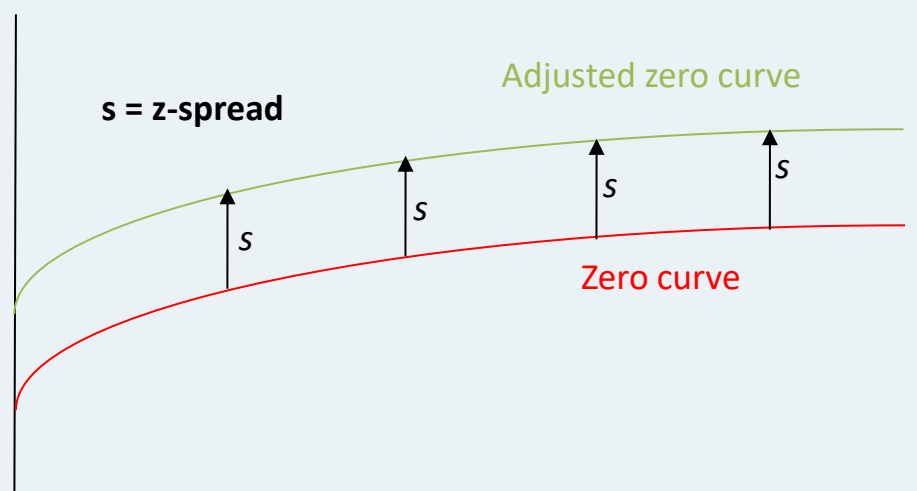
- This  $s$  is known as the **z-spread**

Source: J.P. Morgan

### 3. Z-Spreads – continued

#### Z-Spreads

- The z-spread is a **spread to swaps**, not a spread to government.
  - For this reason z-spreads are normally lower than government spreads as swap rates are generally higher than government yields.
- The z-spread can be thought of as the parallel shift in the zero curve required to match the market observed bond price.
- Advantages:
  - Useful spread measure for people benchmarking themselves to swaps.
  - Takes full structure of bond payments into account.
- Disadvantages:
  - Can't actually “trade” a z-spread.
  - Assumes credit spreads are flat.



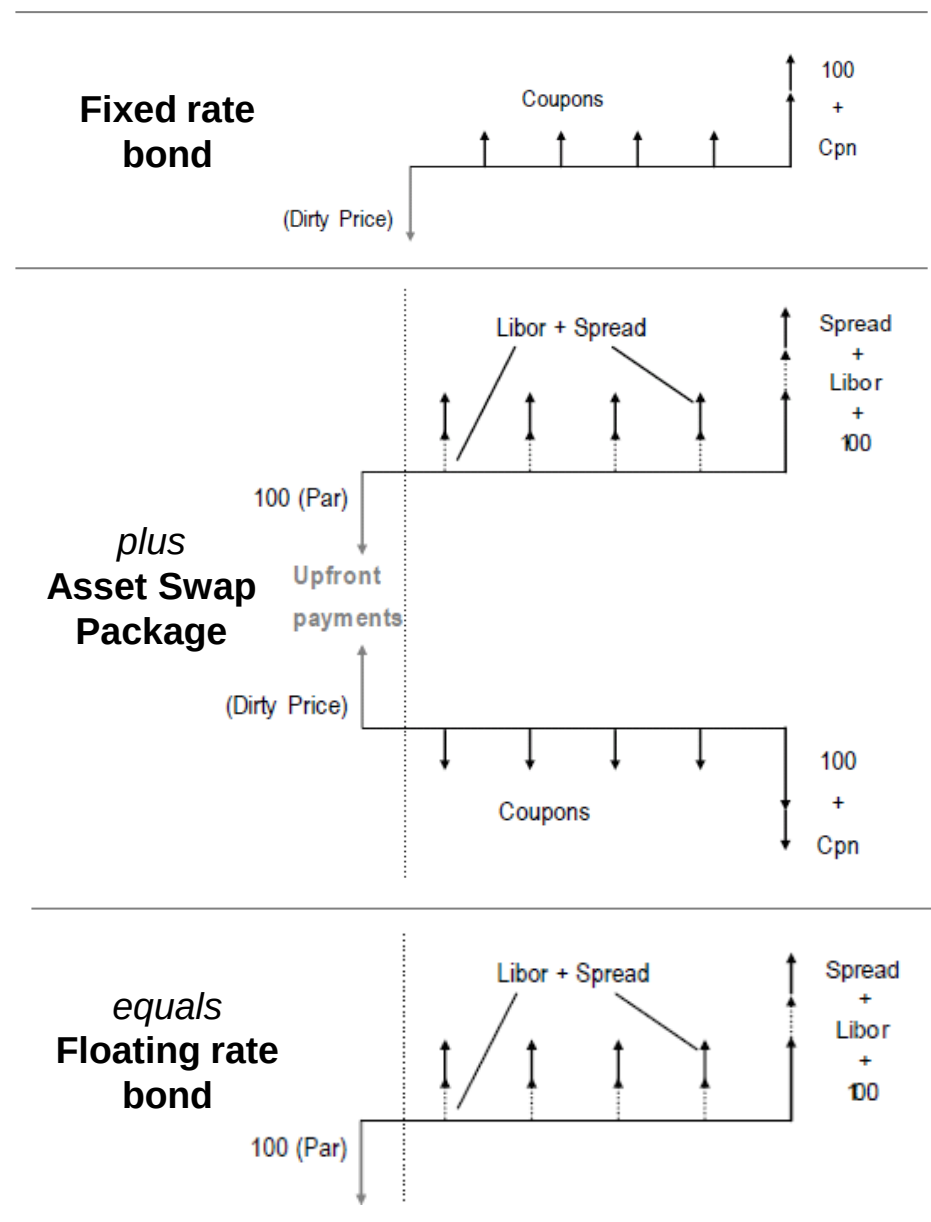
Source: J.P. Morgan

## 4. Asset Swap Spreads

### Asset Swap Spreads

- Asset swaps let investors with fixed rate bonds convert them into bonds paying a floating rate plus a spread.
- This additional spread is known as the asset swap spread.
- The lower the bond price, the higher the asset swap spread.
- Advantages:
  - Unlike z-spreads, asset swap spreads can be traded.
- Disadvantages:
  - Asset swap spreads assume credit risk is flat.
  - Asset swap spreads can understate credit risk for riskier names due to the calculation method.
  - Similarly, they overstate credit risk for bonds trading above par

Source: J.P. Morgan



## 4. Asset Swap Spreads

### Calculation method

- The asset swap package turns the bond into a par-floater, by converting the coupons and the difference between bond's price over par into a running spread over the swap reference rate.
- To calculate the ASW, we need to find the spread over a swap annuity such that the present value of the floating leg equals the present value of the bond coupons plus the difference between par and the bond price:

$$\underbrace{\sum_i^{N_{Float}} (Swap Rate_i + ASW) \Delta_i DF_i}_{\text{Floating Leg PV}} = \underbrace{\sum_i^{N_{Coupon}} C \Delta_i DF_i}_{\text{Coupon Leg PV}} + \underbrace{100 - P}_{\text{Discount vs Par}}$$

- The solution can be approximated closely as:

$$ASW \approx C - IRS_{rate} + \frac{100 - P}{PV01_{IRS}}$$

- This makes intuitive sense:
  - for bonds trading at par the ASW is just the difference between the bond coupon and the interest rate swap rate
  - for bonds trading away from par, the discount 100-P gives rise to an adjustment, annualised via the interest rate swap annuity

Source: J.P. Morgan

## 5. Option-Adjusted Spreads

### OAS

- Some bonds have **embedded optionality**, meaning that the market price of a bond reflects not only the value of the cash flows, but also the value of the embedded option.
- Most commonly, the option gives the issuer the right to **redeem the bond early**.
- For example, take a 5y bullet bond with 2% coupon trading at \$95, a z-spread of 200bp.
- Now, consider a second, otherwise identical bond but which allows the issuer to call early it at 100.
  - **This option is valuable to the issuer:** If the bullet bond rises to a price of 105, the issuer can (and should) redeem the callable bond at 100 and re-finance at a lower coupon. The investor gets 100, but misses out on the 5 point gain he would have realised with the bullet bond.
  - The investor thus effectively sells an option to the issuer which has to be reflected in the bond's price. Let's say this option is worth \$2.5, so the callable bond should have a price of \$92.5.
  - Z-spreads ignore option components so the callable bond will have a wider spread of ~250bp.
  - OAS calculations effectively remove the option component before calculating the bond's spread over a risk-free curve. OAS can be thought of as "option-removed" spread. In this example it should be close to 200bp, i.e. the z-spread of the straight bond.
- OAS calculation mechanics are rather involved and rely on an option pricing and interest rate evolution model. Note that by default BBG calculates OAS to sovereign benchmark curve.

Source: J.P. Morgan



## Spread recap

### Bond spreads

- Corporate bond yields are made up of a number of factors, including credit spreads.
- The higher the default risk, the more spread compensation an investor will demand.
- Two broad types of credit spread for bonds:
  - Spreads based on government bonds:
    - Spread to govt**
    - Interpolated spread to govt**
  - Spreads based on interest rate swaps:
    - Z-spreads**
    - Asset swap spreads**
- Spread used will depend on type of investor and market convention.
  - Typically, US use spread to government, Europe uses z-spread over swaps.
- Here we have only considered fixed coupon bullet bonds. Other spread types exist for floating rate, callable, PIK bonds etc.

Source: J.P. Morgan

# Bloomberg YAS Screen

Spread to government

Interpolated spread to government

Inter'd spread to swaps

Z-Spread

Asset Swap Spread

Option-Adjusted Spread

**YAS 3-BLOOMBERG**

CANCEL HELP SEARCH NEWS QUOTE QUOTE MSG MENU PRINT PG BA PG FW

< > VW 1 1/4 06/10/24 Corp YAS Related Functions Menu Msg: +40

VW 1 1/4 06/10/24 Corp Settings Yield and Spread Analysis

99.237/99.457 1.385/1.346 MSG1 @ 08:20 95 Buy 90 Sell

1) Yield & Spread 2) Graphs 3) Pricing 4) Description 5) Custom

VW 1 1/4 06/10/24 (XS1830986326)

| Spread |                     | Risk               |          | Workout        | OAS   |
|--------|---------------------|--------------------|----------|----------------|-------|
| Spread | 152.63 bp vs        | DBR 1 1/2 05/15/24 |          |                |       |
| Price  | 99.4567             | 109.885            | 14:13:43 | 5.659          | 5.741 |
| Yield  | 1.346003 Wst        | -0.180284          | Ann      | 5.634          | 5.716 |
| Wkout  | 06/10/2024 @ 100.00 | Consensus          | 6 6      | 0.382          | 0.364 |
| Settle | 07/10/18            | 07/10/18           |          | DV 01 on 1MM   | 563   |
|        |                     |                    |          | Benchmark Risk | 6.223 |
|        |                     |                    |          | Risk Hedge     | 905 M |
|        |                     |                    |          | Proceeds Hedge | 904 M |

| Spreads    |       | Yield Calculations |          | Invoice           |            |
|------------|-------|--------------------|----------|-------------------|------------|
| 11) G-Sprd | 151.8 | Street Convention  | 1.346003 | Face              | 1,000 M    |
| 12) I-Sprd | 95.4  | Equiv 2 /Yr        | 1.341504 | Principal         | 994,567.00 |
| 13) Basis  | 12.2  | Mmkt (Act/ 360)    |          | Accrued (32 Days) | 1,095.89   |
| 14) Z-Sprd | 96.5  | True Yield         | 1.345987 | Total (EUR)       | 995,662.89 |
| 15) ASW    | 93.9  | Current Yield      | 1.257    |                   |            |
| 16) OAS    | 143.1 |                    |          |                   |            |

After Tax (Inc 40.800 % CG 23.800 %) 0.812944

Map cross-border equity and bond action. **FOFP**

TYPES OF SPREADS

Source: Bloomberg

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## Spread conventions

### Conventions

- Spreads always quoted in **basis points**
  - 100bp = 1%
- High grade bonds are quoted on a spread basis, not price.
  - Market of **+80/75** means that the trader will buy bonds at a spread of 80 and sell them at a spread of 75.
- Spreads are inversely related to prices – just like yields.
  - If spreads go up, prices go down.
  - If spreads go up by 1bp, then all else remaining constant we expect yields to go up by 1bp.
- Terminology:
  - **Widening**: spreads go up
  - **Tightening**: spreads do down

Source: J.P. Morgan

## What do spreads tell us?

### Spreads

- For single names spread allow us to see how much default risk the market is pricing into the company and how that has changed with time.
- Single name bond spreads can be driven both by company fundamentals and by macro factors.
- We can also calculate average spreads across large numbers of bonds.
- Using these spreads we can get an idea of how much default risk is priced into the overall market.

### AFFP Jul-26 ASW spread



### iBoxx € Corporates ASW Spread



Source: J.P. Morgan, Markit Group

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## Risk in Fixed Income Securities

### Measures of interest rate sensitivity

- In many ways it is easier to measure the risk of fixed income instruments or bonds due to the (for the most part) predictable nature of cash flows.
- The price of a bond is influenced by discount rates, the amount of each cash flow and, crucially, the time over which these cash flows are discounted.

$$\text{Price} = \frac{C}{1+y} + \frac{C}{(1+y)^2} + \frac{C}{(1+y)^3} + \dots + \frac{100+C}{(1+y)^T}$$

- We are interested in the sensitivity to changes in the future discount rates or yields (or spreads).
  - Time is crucial because the change in yield will have a greater effect over longer time periods.
- One measure: weighted average of the times until cash-flow are received:

$$\frac{1 \times C + 2 \times C + 3 \times C + \dots + T \times (100 + C)}{C + C + C + \dots + (100 + C)} = \frac{\sum t_i \times CF_i}{\sum CF_i}$$

- **Duration** replaces the future cash flow with the *present value* of the cash flow.
  - Can be interpreted as an estimate of the bond's price sensitivity to changes in interest rates.
  - It expresses the bond's exposure to interest rate risk.

Source: J.P. Morgan

# Duration

## Modified Duration

- Modified duration tells us how much our bond price changes in % terms for a 1% change in the bond yield.

|            |                           |                                                                        |
|------------|---------------------------|------------------------------------------------------------------------|
| ■ Example: | Bond price                | = 120                                                                  |
|            | Modified duration         | = 10                                                                   |
|            | Yield rises from 4% to 5% |                                                                        |
|            | Change in bond price      | = % yield change x mod duration x bond price<br>= -1% x 10 x 120 = -12 |
|            | <b>New bond price</b>     | = 108                                                                  |

- Mathematically, it is the first derivative of the bond price with respect to the yield

$$\text{Modified Duration} = \frac{1}{\text{Price}} \times \frac{\partial \text{Price}}{\partial \text{Yield}}$$

- The lower the yield, and the lower the coupon, the closer the duration will be to T, the time-to-maturity.
- A 10Y bond will have a duration of ~10 years (a bit less in practice) and from this we can infer the bond would be expected to gain approximately 10% in market value if interest rates fall by 1% (100 basis points) and to lose the same amount if interest rates rise by 1%.
- Modified duration can be used to calculate *the price impact from spread changes* as well – 1bp change in spread has same effect as 1bp change in yield (assuming rates stay constant).
- $DV01 = d\text{Price}/d\text{Yield} \times \text{Notional} = - \text{ModDur} \times \text{Price} \times \text{Notional}$  (scaled appropriately for 1bp impact).

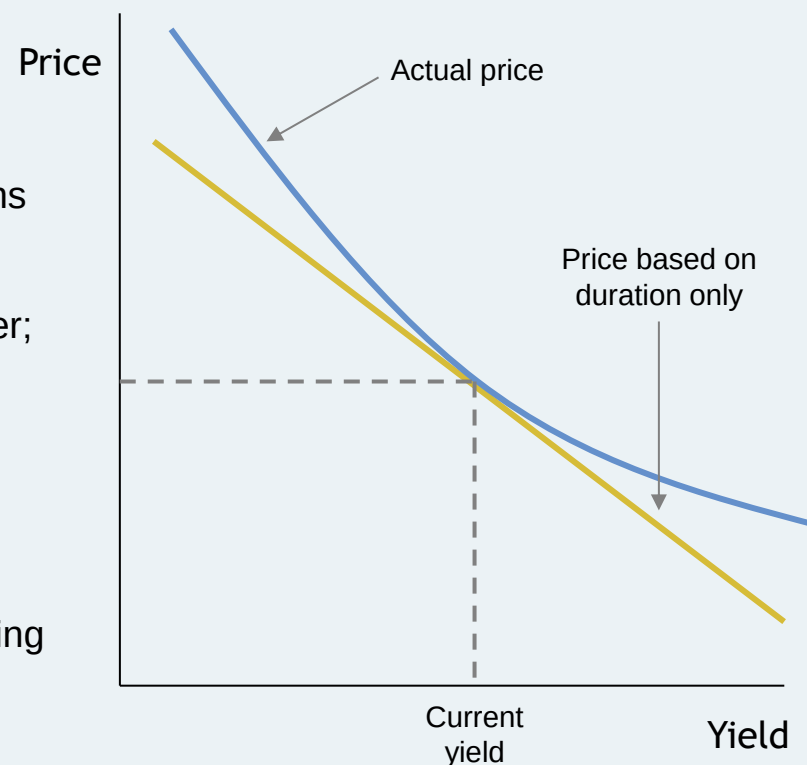
Source: J.P. Morgan



## Bond convexity

### Convexity

- Since duration assumes there is a linear relationship, using modified duration to calculate price changes is only accurate for small moves.
- Durations change as yields change. This is known as **convexity** and means that the relationship between prices and yields is not a simple straight line.
- As yields increase and bond prices fall, durations decrease.
- As yields decrease and bond prices increase, durations increase.
- This makes bond losses smaller and bond gains bigger; bonds have **positive convexity**.
- Simple way to remember it – bond prices cannot go below zero however high yields go, so the bond price curve must flatten out.
- Duration based approximation can be improved by using a second order approximation, convexity.



Source: J.P. Morgan

## Marking bonds to market

### Calculating bond return / P&L

- Two ways to mark bonds to market:
- **1. Calculating bond P&L with prices (exact):**
  - Bond P&L = New Clean Price – Old Clean Price + Carry
  - Carry = Coupon x Day Count Fraction (e.g. ACT/365)
  - Example: buy bond at 95, sell bond six months later at 98. Coupon = 5%
  - P&L = 98 – 95 + 5 / 2 = 5.5% of par notional

- **2. Calculating bond P&L with yield (approximation):**

$$\Delta P = -\text{ModDur} \times P \times \Delta y + y \times P \times \Delta t$$

- Bond P&L (%) = Modified Duration x (Initial Yield – New Yield) + Carry
- Carry = Initial Yield x Day Count Fraction
- Note difference in carry term – we need to use the yield here rather than the coupon as the bond price will change over time even if yield stays the same – this is known as “pull to par”.
- Example: buy bond at yield of 9%, sell bond six months later at yield of 8%. Modified dur = 4.
- P&L (%) = 4 x (9% - 8%) + 9% x 6/12 = 8.5% (of initial price).
- Assume initial price was \$90. Trade size \$100mm of par notional.
- P&L (\$) = 8.5% x \$90 / \$100 x \$100mm = \$7,650,000

## Bloomberg YAS for bond risk metrics

### YAS

|                                      |                                       |                    |                           |                |
|--------------------------------------|---------------------------------------|--------------------|---------------------------|----------------|
| F 3 1/4 02/12/32                     |                                       | \$↑84.437          | -.137                     | Yld .000       |
| At 12:21                             |                                       | Source IBVL        |                           |                |
| F 3 1/4 02/12/32 Corp                |                                       | Settings ▾         | Yield and Spread Analysis |                |
| 84.290/84.560                        |                                       | 6.187/6.131        | IBVL @ 12:11              | Notes          |
| 1) Yield & Spread                    |                                       | 2) Graphs          | 3) Pricing                | 4) Description |
| 5) Custom                            |                                       | 6) Calls           |                           |                |
| F 3 1/4 02/12/32 ( 345370DA5 )       |                                       | Risk               |                           |                |
| Spread                               | 177.54 bp vs 10yT 4 1/4 05/15/35 ▾    | Workout            | OAS                       |                |
| Price                                | 84.56 ↺ 99-05 12:11:37                | ● M.Dur ● Dur      | 5.659                     | 5.687          |
| Yield                                | 6.1310 Wst ▾ 4.3556 S/A ▾             | Risk               | 4.860                     | 4.884          |
| Wkout                                | 02/12/2032 @ 100.00 Consensus Yld 4 4 | Convexity          | 0.376                     | 0.378          |
| Settle                               | 07/08/25 📅 07/08/25 📅                 | DV ▾ 01 on 1MM     | 486                       | 488            |
|                                      |                                       | Benchmark Risk     | 7.904                     | 7.988          |
|                                      |                                       | Risk Hedge         | 615 M                     | 611 M          |
|                                      |                                       | Proceeds Hedge     | 861 M                     |                |
| ✍ Spreads                            |                                       | Invoice            |                           |                |
| Yield Calculations                   |                                       |                    |                           |                |
| 11) G-Sprd                           | 204.0 Street Convention 6.1310        | Face               | 1,000 M                   |                |
| 12) I-Sprd                           | 252.3 Equiv 1 ▾ /Yr 6.2250            | Principal          | 845,600.00                |                |
| 13) Basis                            | -39.8 Mmkt (Act/360 ▾ )               | Accrued (146 Days) | 13,180.56                 |                |
| 14) Z-Sprd                           | 246.7 True Yield ▾ 6.1308             | Total (USD)        | 858,780.56                |                |
| 15) ASW                              | 223.2 Current Yield 3.843             |                    |                           |                |
| 16) OAS                              | 202.5                                 |                    |                           |                |
| After Tax (Inc 40.800 % CG 23.800 %) |                                       | 4.1321             |                           |                |
| Issue Price = 100.000.               |                                       |                    |                           |                |

Source: J.P. Morgan

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# Introducing Credit Default Swaps

October 2025

**Dmytro Shelukhin** <sup>AC</sup>

Credit Index & Derivative Strategy

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# Agenda

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# What is a Credit Default Swap?

## Credit Default Swaps

- Initial rationale:
  - You own a bond of company XYZ
  - If the company defaults you will suffer losses:  $\text{Notional} \times (1 - R)$ 
    - R: Recovery rate (price of the bond, after default, as % of notional)
  - How can you hedge those losses?
    - Selling the bond, or...
    - Buying a financial product which compensates you for the bond losses if default occurs
- **In some cases it is not possible or desirable to sell the bond.**
- Credit Default Swaps (CDS) acts like insurance for a bond or loan – if the bond defaults then CDS pays out  $(1-R) \times \text{Notional}$  to cover bond investors' losses.
- In return for this insurance the bond holder must make regular payments.
- CDS market originated in 1990s from need of banks to hedge loan risk of major clients.
- The market has gone through a four-decade evolution, moving from non-standardized bilateral contracts in the late 1990s to instruments with standardized terms, trading and default settlement conventions today, applicable globally.

Source: J.P. Morgan

# Credit Default Swap Mechanics

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## Credit Default Swaps

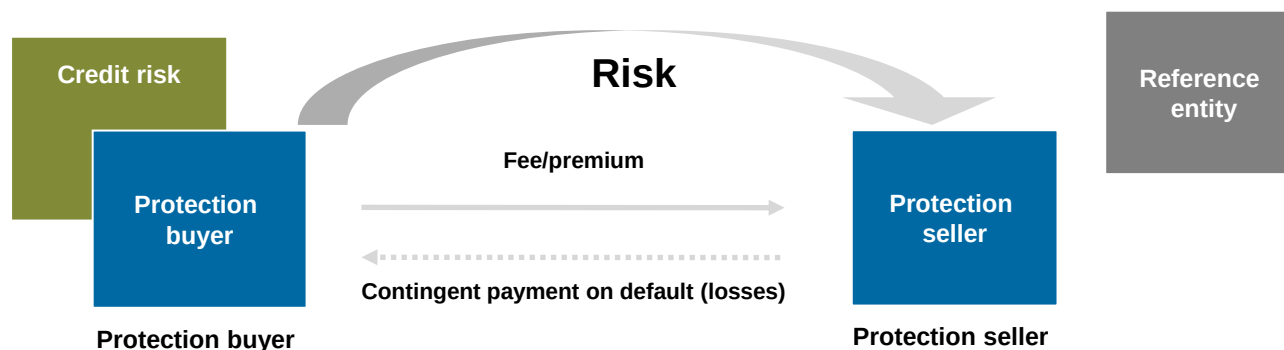
- There are two counterparties in a CDS contract:
  - Buyer of protection
  - Seller of protection
- The protection buyer pays the protection seller regular fees – known as the **spread**.
- In return, if the underlying company defaults, the protection seller will make a payment to the protection buyer equal to  $(1 - \text{Bond Recovery}) \times \text{Notional}$ .
- Each CDS contract specifies (among other things):
  - The underlying company that insurance is being bought on – the “reference entity”.
  - The maturity of the contract – norm is 5 years.
  - Notional of the trade.
  - The contract spread and how this is paid, “full running” or “fixed coupon + upfront” – more later.
  - How we define a default.

Source: J.P. Morgan

# Credit Default Swap Mechanics

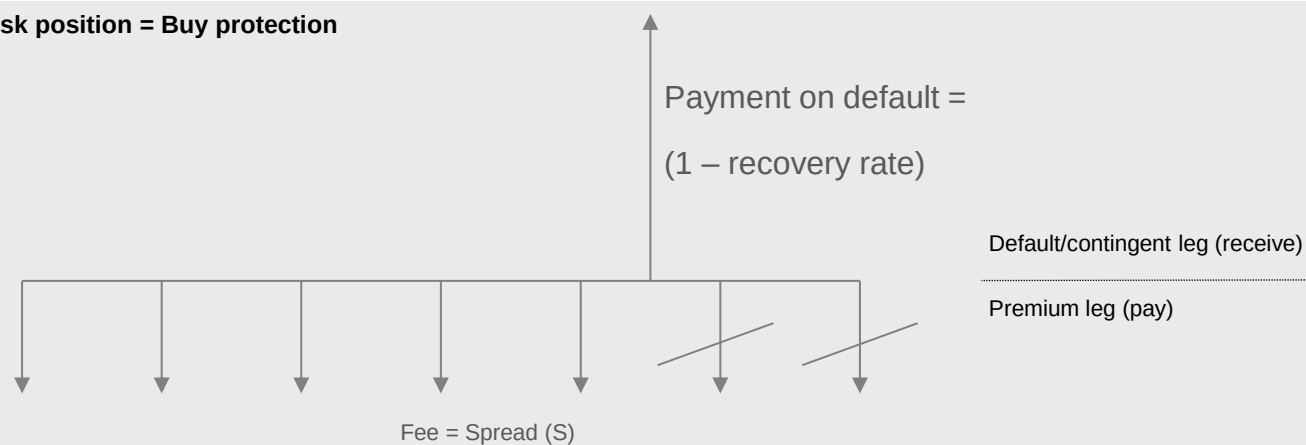
Short risk/buy protection

Long risk/sell protection



## In cash flow terms

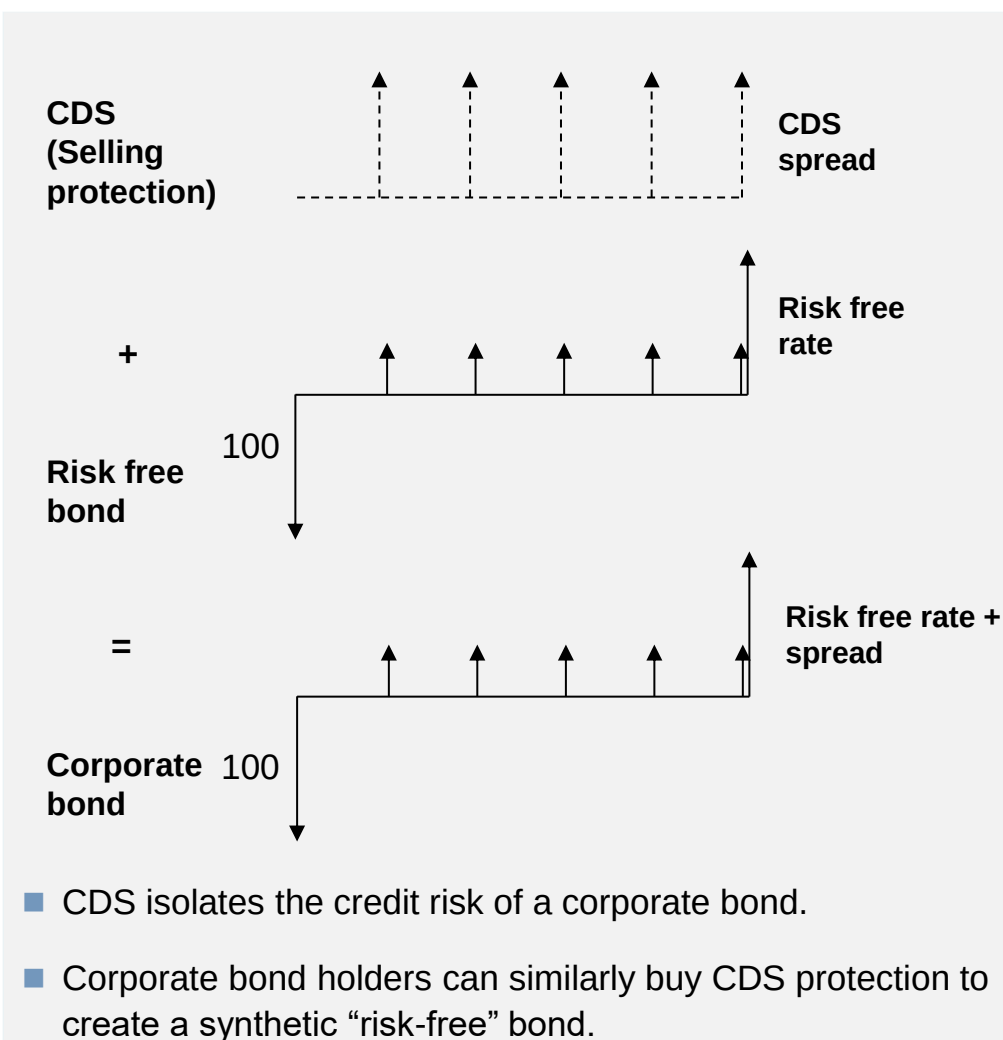
Example of a short risk position = Buy protection



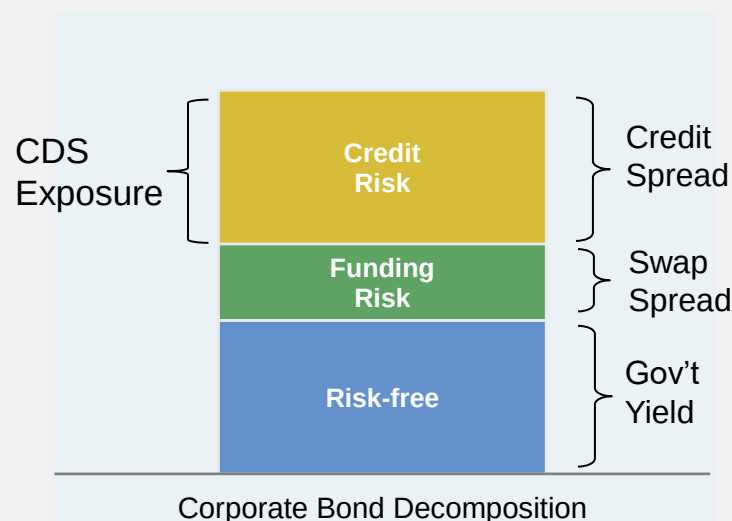
Source: J.P. Morgan

# Credit Default Swap Mechanics – Bond and CDS Comparison

## Credit Default Swaps



- Bond all-in yield can be decomposed into three parts:
  - Credit risk – Credit spread
  - Funding risk – Swap spread
  - Risk-free rate – Treasury rate
- Investors who only want to trade credit risk and not take exposure to government yields or swap rates can find credit default swaps more suited to their needs.



# Agenda

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## CDS Contracts in Practice

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### Standardised contracts

- Credit default swaps are “over-the-counter” (OTC) contracts.
  - Bilateral contracts between two counterparties
  - Counterparties can use whatever terms and conditions they want
  - Counterparties take on each other’s credit risk, mitigated by central clearing
- In practice:
  - Market uses the same “standard” contract
  - Standard contract uses ISDA 2014 Credit Derivative Definitions
  - Standard contract terms updated in 2009 – known as “Big Bang” and “Small Bang”
  - CDS Auction settlement is hardwired, allowing both cash and physical settlement
  - New CDS definitions came in effect in September 2014
  - Standardisation greatly increased volumes in CDS
  - CDS contracts are collateralised on a daily basis and are increasingly centrally cleared, reducing counterparty risk.

Source: J.P. Morgan

## CDS Contract Example

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### CDS Contract

|                                   |                                                                 |
|-----------------------------------|-----------------------------------------------------------------|
| ■ <b>Counterparties:</b>          | JPM (seller of protection)<br>ABC Capital (buyer of protection) |
| ■ <b>Reference Entity:</b>        | XYZ Ltd.                                                        |
| ■ <b>Maturity:</b>                | 5 years                                                         |
| ■ <b>Notional:</b>                | €10 million                                                     |
| ■ <b>CDS Spread:</b>              | 3% per annum, ACT/360                                           |
| ■ <b>Payment Frequency:</b>       | Quarterly: March, June, September, December                     |
| ■ <b>Credit Events:</b>           | Bankruptcy, Failure to pay, Restructuring                       |
| ■ <b>Obligations:</b>             | Borrowed money                                                  |
| ■ <b>Settlement:</b>              | Cash or Physical; via a CDS auction                             |
| ■ <b>Deliverable Obligations:</b> | Set of characteristics.                                         |
| ■ <b>Reference Obligation:</b>    | Specific bond (ISIN)                                            |

Source: J.P. Morgan

## CDS Maturities and Size

---

### CDS Maturities

- CDS contracts are traded with standardised maturities based on semi-annual IMM dates:
  - 20<sup>th</sup> June and December of each year.
- The 5y maturity is the most liquid point. Currently the active 5y maturity is the 20 June 2029; on 20<sup>th</sup> September we will “roll” to a new contract and the 20 Dec 2029 contract will become the new liquid 5y point.
- All points in the curve are traded – 3M, 6M, 1Y, 2Y...10Y – but 75%+ volume is at 5y point.

### CDS Trade Sizes

- Typical investment grade CDS trade size: \$5mm to \$20mm
- Typical high yield CDS trade size: \$2mm to \$10mm.
- In many cases single name CDS is more liquid than corporate bonds – there can be many bonds from one company but only one standardised 5y CDS contract.

Source: J.P. Morgan

# Credit Events and Obligations

## Credit Events

- For a CDS contract to trigger a “credit event” must occur.
- Credit events fall into three main types.
  - **Bankruptcy** – covers bankruptcy filings, insolvency, appointment of administrators/liquidators for the reference entity.
  - **Failure to pay** – when the reference entity misses a coupon or principal payment on any *obligation*, taking a grace period into account.
  - **Restructuring** – a change to specific bond terms as a result of a credit deterioration. Includes reduction of principle, reduction of coupon, change to maturity, currency or seniority.
    - North American CDS contracts do not have restructuring as a separate credit event; companies usually enter Chapter 11 bankruptcy instead.
  - **Governmental Intervention (just for European Banks)** – Bail-in scenario.
- EU sovereigns have different credit events: Failure To Pay, Restructuring, Repudiation/Moratorium.
- Credit events are only triggered by actions on defined **Obligations**
  - Obligations typically cover “borrowed money” – e.g. bonds, loans, deposits.
  - Trade payables are not borrowed money.
  - Non-standard CDS contracts can have different definitions of obligations – e.g. a single bond.

Source: J.P. Morgan

# Agenda

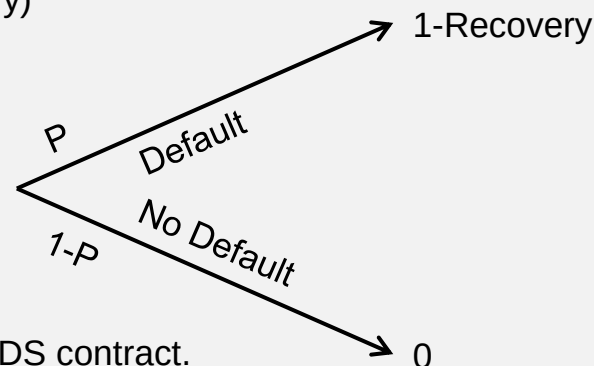
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## CDS Spreads and Default Probabilities

### Default probabilities

- The higher the probability that a company defaults, the higher the spread that the buyer of protection will pay to enter into a CDS contract.
- What is the exact relationship between default probability and spread?
- Take a 1y CDS contract.
  - Company has  $P$  probability of default of having defaulted at the end of the year.
  - If company defaults, buyer of protection receives (1-Recovery)
  - If company survives, buyer of protection receives nothing.
- We can use these probabilities to calculate the expected payoff from the CDS contract.
  - Expected payoff =  $P \times (1-R) + (1-P) \times 0$   
=  $P \times (1-R)$
- The buyer of protection expects to receive  $P \times (1-R)$  from the CDS contract.
- For the contract to be fairly priced, the spread paid by the buyer of protection must equal the expected payoff.
- **Spread = Probability of Default x (1 – Recovery)**



## CDS Spreads and Default Probabilities - Continued

### Default probabilities

- **Spread = Probability of Default x (1 – Recovery)**
- For example, if the 1y default probability is 2% and the recovery is 40% the CDS spread will be equal to 120bp ( $=2\% \times (1-40\%)$ ).
- In practice, the probabilities of default are not known but we use this relationship to back them out from market traded spreads and assumed recovery rates.

### **Implied Probability of Default = Traded Spread / (1 – Recovery)**

- This relationship also applies for CDS contracts with maturities of longer than 1Y, but in this case the implied probability of default is the implied *annual* probability of default.
- Example:
  - 5y CDS contract
  - Spread = 500bp
  - Recovery = 40%
- Annual probability of default = 8.33% ( $=500\text{bp} / (1-0.4)$ )
- In reality, the relationship between default probabilities and spreads is more complicated and takes into account factors such as discounting, but the above relationship is a good approximation.

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## Four Parameters Uniquely Define a Credit Default Swap

---

### ■ **Which Credit** (Note: Not which bond, but which company)

- Credit default swap contracts specify a reference obligation that
  - (1) robustly defines the issuing entity through the bond prospectus and
  - (2) indicates what level of the capital structure is delivered in default (typically senior unsecured bond).

### ■ **Notional Amount**

- Amount of credit risk being transferred
- Agreed between the buyer and seller of CDS protection.

### ■ **Spread** (or Upfront / Price)

- No matter the quoting convention, two cash flows are agreed on:
  - (1) the upfront amount and
  - (2) the annual fixed coupon (100bp or 500bp), where payments are paid quarterly, and accrue on an Actual/360 day basis.

### ■ **Maturity**

- The expiration of the contract, usually on the 20th of March, June, September or December.
- Generally, the 5-year contract is the most liquid.

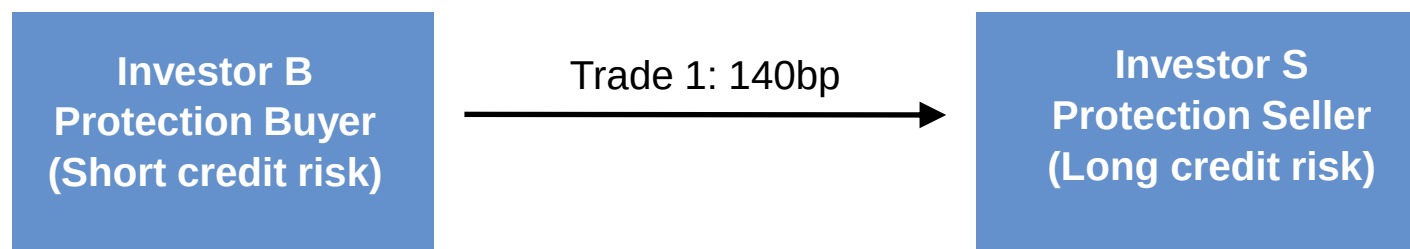
## Example: Buying CDS protection (short risk) on Ford

### Ford trade

- If investor B believes Ford's creditworthiness will worsen, they would buy CDS protection (short credit risk), paying 140bp annually, in our example
- If spreads on Ford increase ( $>140\text{bp}$ ) this benefits the investor who has bought protection
- If they decrease ( $<140\text{bp}$ ) it causes the investor who has bought protection to lose money

### Trade 1: Today

|             |                                                       |
|-------------|-------------------------------------------------------|
| Investor B: | Buy CDS protection<br>(short credit risk, pay spread) |
| Credit:     | Ford Motor Co                                         |
| Notional:   | \$10mm                                                |
| Spread:     | 140bp                                                 |
| Maturity:   | 5 years (20/12/2030)                                  |



Source: J.P. Morgan

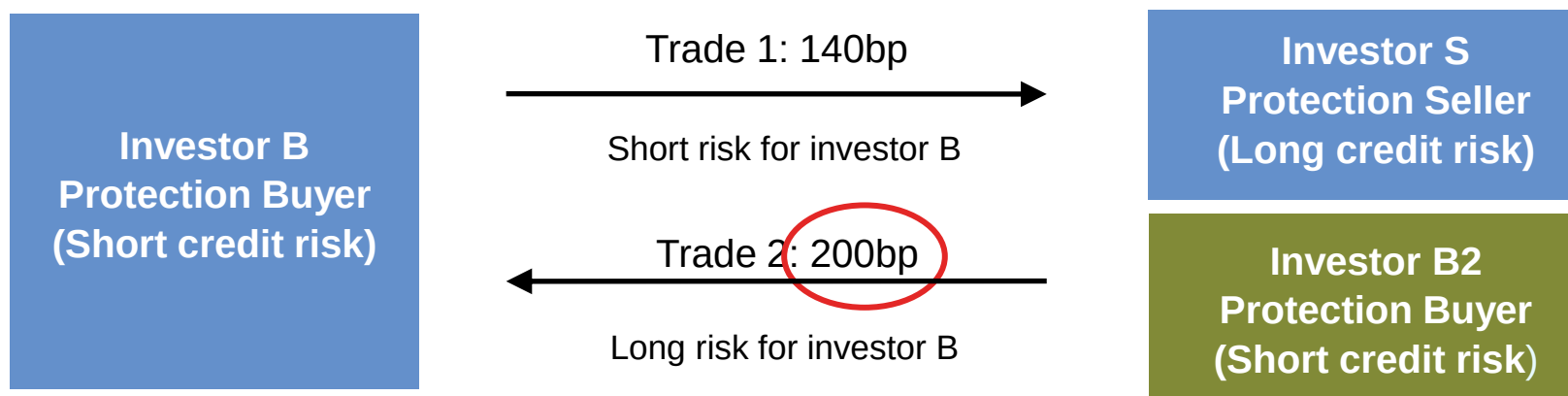
## If Ford's Credit Worsens: Monetize Gains by Entering into Opposite Trade

### Exiting the trade 6 months later

- Assume that, after six months, Ford's spread widens by 60bp to 200bp ( = 140bp + 60bp)
- Investor B could enter into an opposite trade, namely sell CDS and receive 200bp annually. They continue to pay 140bp annually, so net 60bp per year until 2030.
- If Ford defaults, investor B will stop receiving the annual net 60bp. The investor has a "default neutral" position in the sense that, if there is a default, the gain in one trade (short risk) is the same as the loss in the other (long risk)

### Trade 2: Unwind in 6 month

|             |                                           |
|-------------|-------------------------------------------|
| Investor B: | Sell CDS protection<br>(long credit risk) |
| Credit:     | Ford Motor Co                             |
| Notional:   | \$10mm                                    |
| Spread:     | 200bp                                     |
| Maturity:   | 4.5 yrs (20/12/2030)                      |



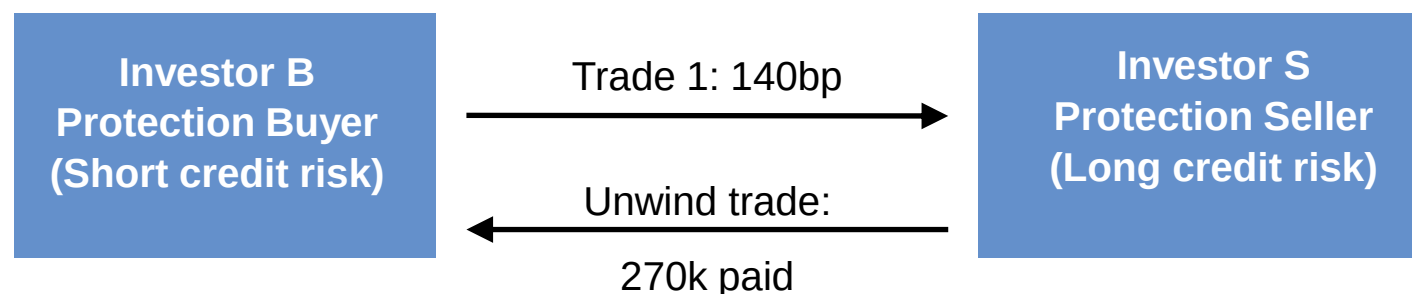
Source: J.P. Morgan

Net annual payment to investor B = 60bp

J.P.Morgan

## The second, more common method to monetize trades is to unwind them

- Investor B can unwind Trade 1 with Investor S, or with another dealer, presumably at a better price.
- Investor B receives from Investor S cash equal to the present value of the expected payments, or an unwind payment. The present value calculation incorporates the probability of Ford's default
- The unwind payment is the PV of 60bp until the maturity of the contract:  
(change in spread) x (risky annuity of par CDS) x (notional)  
= 60bp x 4.5 x \$10mm ≈ \$270k
- After unwinding the trade, investor B has no outstanding positions



Source: J.P. Morgan

## What is the value of 1bp?

### Calculating the Risky Annuity (aka RPV01)

- We need to work out what the present value of the 60bp annual payment is to calculate our mark-to-market. We can do this by calculating the present value of a 1bp annual cashflow over the next five years – known as an annuity.
- **Example: a 1bp annuity for 5y.** We will receive 1bp every year for the next five years.
- If interest rates and default risk is zero, the PV of this annuity is worth 5.
- With non-zero interest rates, the value of later payments decreases. The new PV is 4.7.
- **What happens if the company defaults before the end of 5y?**
- If there is a default in the reference entity, we stop receiving the net 60bp coupon every year. Therefore, the probability of receiving each 60bp payment is less than 100% and is linked to the riskiness (survival probability) of the company.
- We need to take the possibility that the company might default and we might not receive later payments into account when calculating the PV of this annuity.
- When credit risk is taken into account (the risk of not actually receiving some payments), the expected value of later payments decreases further. The new PV is 4.5.

This PV is called the **risky annuity** and is very important in valuing CDS contracts.

Source: J.P. Morgan

## Risky Annuity

What is the present value of an annual 60bp cashflow lasting 5y?

### Non-Risky Zero Interest Rate World

$$PV = 60bp \times 5$$

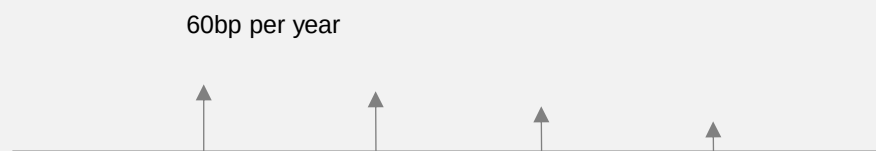
$$PV = 300c = 3\%$$



### Non-Risky Non-Zero Interest Rate World

$$PV = 60bp \times 4.7$$

$$PV = 282c = 2.82\%$$



### Risky Non-Zero Interest Rate World

$$PV = 60bp \times 4.5$$

$$PV = 270c = 2.7\%$$



Source: J.P. Morgan.

## Calculating the Mark-to-Market on Full Running Spread trades

### Ford MTM

- For our Ford trade we bought protection at 140bp and sold protection at 200bp six months later.
- We are receiving net 60bp from this trade.
- To mark this trade to market we need to calculate the present value of this 60bp.

|                  |   |                                                         |
|------------------|---|---------------------------------------------------------|
| ■ Mark to market | = | Risky Annuity x (Exit Spread – Entry Spread) x Notional |
|                  | = | 4.5 x (200bp – 140bp) x \$10mm                          |
|                  | = | \$270,000                                               |

- This 60bp net coupon is worth \$270,000 to us today.
- To calculate full P&L for this trade we also need to calculate the carry we have paid up until now.
- We paid 140bp for six months on a notional of \$10million.

|         |   |                                         |
|---------|---|-----------------------------------------|
| ■ Carry | = | Entry Spread x Time in Years x Notional |
|         | = | 140bp x 0.5 x \$10mm                    |
|         | = | \$70,000                                |

|             |   |                               |
|-------------|---|-------------------------------|
| ■ Total P&L | = | Mark-to-market – Carry        |
|             | = | \$200,000 (or 2% of notional) |

Source: J.P. Morgan

## Annuity Risk in Full Running Spread trades

### Annuity Risk

- We have exited our long protection position in Ford by selling protection on the same notional.
  - We receive \$60,000 every year from the difference in the two spreads.
  - The present value of these payments gives us a mark-to-market profit of \$270,000
- What happens if the company defaults tomorrow?
  - All coupons would cease and we wouldn't receive any of our \$60,000 payments.
  - Our positive mark-to-market of \$270,000 would disappear.
  - Our position still has default risk despite us exiting the trade!
  - This is known as **annuity risk**.
- Even if the company does not default, we still have exposure to the default risk of the company. We value our 60bp annual payment using the risky annuity; the risky annuity will fall as spreads widen and default becomes more likely.
- What's the solution? Trading CDS on **Fixed Coupons + Upfronts** instead of full running spreads.

Source: J.P. Morgan



## Fixed Coupon + Upfront Trading Convention

### Practically, CDS trades with upfront and fixed coupon

- Since 2009, nearly all CDS contracts trade with a **Fixed Coupon + Upfront** format rather than Full Running Spread.
- Standard coupons = 100bp for IG names, 500bp for HY names.
- In our original Ford trade, we bought protection at 140bp.
  - Full Running Spread format – we pay 140bp every year.
  - **Fixed Coupon format – we pay standard 100bp coupon every year.**
  - We are only paying 100bp for something worth 140bp!
  - **To compensate for this we also pay a one-off upfront amount.** This is equal to the present value of the difference between the 140bp we should be paying and the 100bp we are actually paying.
- To calculate the upfront we should pay, we need to calculate the present value of what we should be paying (140bp) and what we are actually paying (100bp). We can do this using the risky annuity we discussed in the previous section.

$$\text{Upfront} = \text{Risky Annuity} \times (\text{Quoted Spread} - \text{Coupon}) \times \text{Notional}$$

- If the quoted spread is higher than the coupon then the buyer of protection will pay the upfront. If the quoted spread is lower than the coupon then the seller of protection will pay the upfront.
- For our Ford example, the buyer of protection will pay \$180,000 ( $= 4.5 \times (140 - 100) \times \$10\text{mm}$ )

Source: J.P. Morgan

## Trading Ford with Fixed Coupons and Upfronts

### Fixed Coupon + Upfronts

- We carry out the same Ford trade as before, but use an upfront + fixed coupon format rather than a full running spread format this time.
- We buy 5y protection at 140bp with a fixed coupon of 100bp on notional of \$10mm.
  - $\text{Upfront} = 5 \times (140\text{bp} - 100\text{bp}) \times \$10\text{mm} = \$200\text{k}$ .
  - We pay upfront of \$200k and pay coupon payments of \$100k every year.
- Six months later, we sell protection at 200bp with a fixed coupon of 100bp on notional of \$10mm.
  - $\text{Upfront} = 4.5 \times (200\text{bp} - 100\text{bp}) \times \$10\text{mm} = \$450\text{k}$ .
  - We receive upfront of \$450k and receive coupon payments of \$100k every year.
- Mark to market
  - = New upfront – Old upfront
  - = \$450,000 - \$200,000
  - = \$250,000
- We have paid six months of carry equal to \$50,000 ( $= 100\text{bp} \times 0.5 \times \$10\text{mm}$ ).
- Total P&L
  - = Mark to market – Carry
  - = \$250,000 – \$50,000
  - = \$200,000

Source: J.P. Morgan

## Annuity Risk in Fixed Coupon + Upfront Trades

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### Closing Out Fixed Coupon Trades

- After closing out trade we will:
  - Pay 100bp coupon in original contract.
  - Receive 100bp coupon in second contract.
  - Net we have no ongoing payments.
- If the company defaults we have no exposure as we are not receiving any net payments.
- **There is no annuity risk in Fixed Coupon + Upfront trades** – as long as trades use the same fixed coupon.

Source: J.P. Morgan

## Why is there an upfront and how large is it?

### Upfront & Fixed Coupon Trades

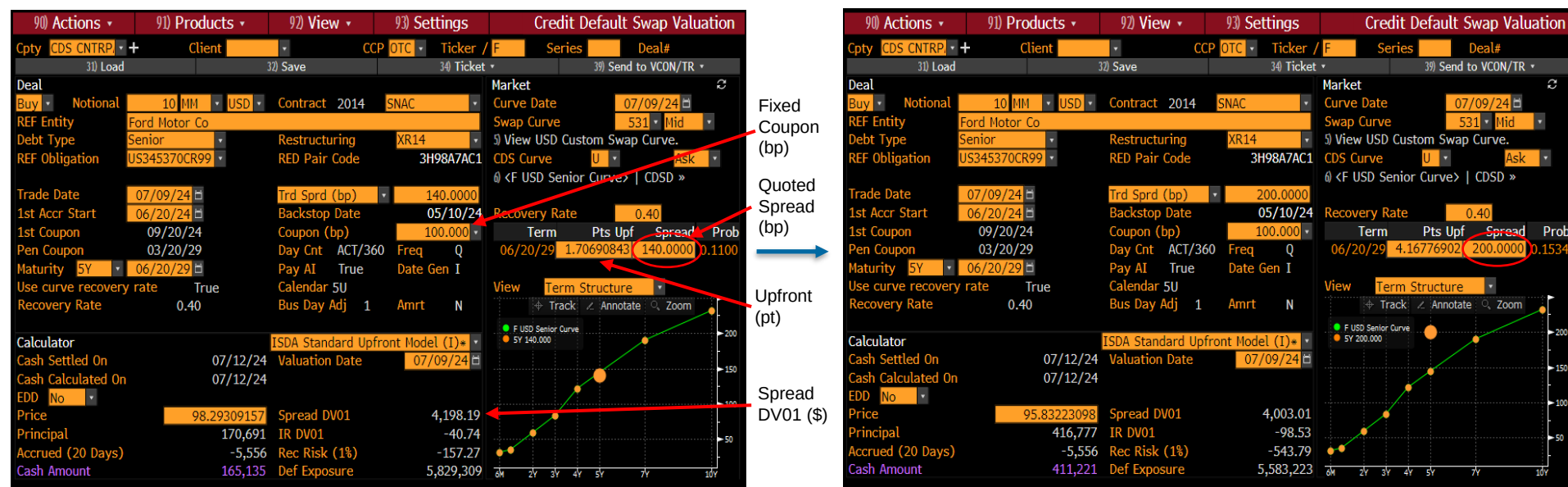
- Makes all contracts completely fungible.
- Because coupon is fixed and credit risk might be larger or smaller than the coupon.
- If spread is higher than coupon then coupon is too small for actual risk. Therefore, you need to pay an upfront amount to make up for this.
- Similar to bonds trading at a discount, with yield higher than coupon.
- Upfront is really like clean price of a bond (more precisely,  $\$100 - \$UF$  is CDS in bond clean price terms).
- Assume 5y CDS market spread is 140bp: For a 100bp contract, upfront is 1.8%.
- \$1.8 is the present value of the 40bp you are not paying each year on your \$100 protection (40bp = 140bp - 100bp coupon); therefore, protection buyer pays 1.8% upfront.

|                         | 140bp coupon | 100bp coupon |
|-------------------------|--------------|--------------|
| Today                   | 0            | \$1.8        |
| Year 1                  | \$1.4        | \$1.0        |
| Year 2                  | \$1.4        | \$1.0        |
| Year 3                  | \$1.4        | \$1.0        |
| Year 4                  | \$1.4        | \$1.0        |
| Year 5                  | \$1.4        | \$1.0        |
| <b>Total paid</b>       | <b>\$7</b>   | <b>\$6.8</b> |
| <b>PV of total paid</b> | <b>\$6.5</b> | <b>\$6.5</b> |

Source: J.P. Morgan

# Working out the Ford PnL Example Using CDSW

## ISDA CDS Standard Model aka CDSW Upfront-Spread Converter in Bloomberg



- 140bp spread with 100bp coupon assuming a recovery rate of 40% = 1.71% upfront
- 200bp spread with 100bp coupon assuming a recovery rate of 40% = 4.17% upfront
- (Instantaneous) PnL = 4.17% - 1.71% = 2.46% or \$246k on \$10mn notional trade
- Approximate PnL with Spread Change x Spread DV01 = (200bp – 140bp) x 4.2 = 2.52%
- Bloomberg Excel Add-in supports ISDA CDS Upfront/Price/Spread Converter:
  - =BDP("ITRX MAIN CDSI S43 5Y Corp","CDS\_QUOTED\_PRICE","CDS\_FLAT\_SPREAD",140,"SETTLE\_DT","20250709","SW\_CURVE\_DT","20250709")

# Agenda

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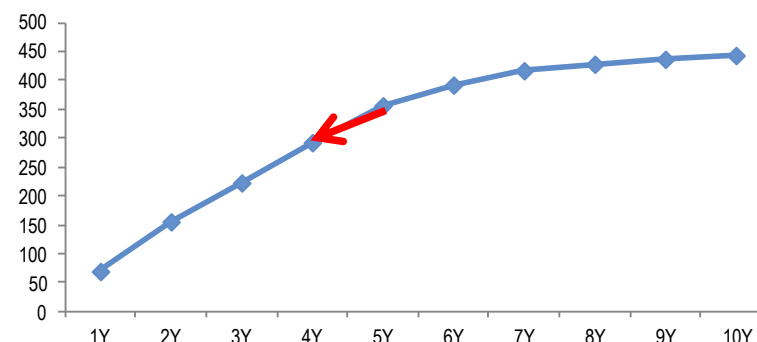
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# Carry and Slide

## Carry, Slide and Time Value

- CDS curves give us an idea of how the spreads of CDS trades could behave in the future (based on the current pricing).
- Many investors base future P&L projections on the idea that CDS spreads will “slide” down (or up) the CDS curve.
- For example, if we sell protection on a 5y CDS contract at a spread of 350bp and the 4y CDS spread is 300bp, then we can calculate the P&L impact of the spread tightening to 300bp over the next year on our position.
- This projected P&L impact is known as **slide**.
- Slide is not a guarantee of performance; spreads can clearly rise or fall over the next year. It does, however, give investors an idea of how much the rolldown on the curve is worth assuming that the shape of the curve remains constant over time.
- **Carry** is the P&L we get based on coupons being paid over time. Together, carry and slide make up **time value**, which is the total projected P&L of the position over time.
- Carry and Slide are calculated differently depending on whether the trade is in a **full running spread** or **fixed coupon + upfront** convention.

## Rolldown from 5y to 4y



## Carry and Slide for Full Running Spread Trades

- We sell 5y protection on a full running spread format on a name with the following par CDS curve:

| Tenor | Spread | Risky Annuity |
|-------|--------|---------------|
| 1Y    | 80bp   | 0.9           |
| 2Y    | 95bp   | 1.8           |
| 3Y    | 110bp  | 2.7           |
| 4Y    | 130bp  | 3.6           |
| 5Y    | 150bp  | 4.5           |

- Every year we will receive a coupon of 150bp.
  - **Carry over next 12m = 5y Spread = 1.50%**
- After one year we expect the 5Y spread to roll down to 130bp. The slide is equal to the mark-to-market resulting from this spread move.
  - **Slide over next 12m = (5y Spread – 4y Spread) x 4y Risky Annuity = (150bp – 130bp) x 3.6 = 0.72%**
- **Time Value = Carry + Slide = 2.22%**
- If we had bought 5y protection the rolldown in the curve would work against us, so the time value would be -2.22%.



## Carry and Slide for Fixed Coupon + Upfront Trades

- We sell 5y protection on a fixed coupon plus upfront format on a name with the following par CDS curve with a coupon of 100bp:

| Tenor | Spread | Risky Annuity |
|-------|--------|---------------|
| 4Y    | 130bp  | 3.6           |
| 5Y    | 150bp  | 4.5           |

- Every year we will receive a coupon of 100bp.
  - **Carry over next 12m = Fixed Coupon = 1.00%.**
  - Note this is different to the Full Running Spread case, where the carry was equal to the 5y Spread.
- When we initially sell protection, the upfront is equal to:  
 $5y \text{ Upfront} = 5y \text{ Risky Annuity} \times (5y \text{ Spread} - \text{Coupon}) = 4.5 \times (150bp - 100bp) = 2.25\%$
- One year later we expect the upfront we unwind the trade at to be equal to:  
 $4y \text{ Upfront} = 4y \text{ Risky Annuity} \times (4y \text{ Spread} - \text{Coupon}) = 3.6 \times (130bp - 100bp) = 1.08\%$
- The upfront has decreased by 1.17% over the year, resulting in a mark-to-market gain of 1.17% in our favour. Some of this change in upfront is due to the change in maturity and some is due to the change in spread; we want to separate the two impacts.
- If the spread had stayed constant at 150bp, the upfront would still have changed due to the shorter maturity. We assume that the 4y risky annuity would be equal to 3.5 if this spread had remained constant. This would have resulted in a new 4y upfront of 1.75%. This means the upfront would have reduced by 0.50% even if the spread had remained constant. We call this the **Excess Carry** ("pull to par").
  - **Excess Carry over next 12m = 0.50%.**
- The slide is the change in the upfront not explained by the excess carry.
  - **Slide over next 12m = 5y Upfront - 4y Upfront - Excess Carry = 2.25% - 1.08% - 0.50% = 0.67%**
- **Time Value = Carry + MtM change = Carry + Excess Carry + Slide = 1.00% + 0.50% + 0.67% = 2.17%**
- Note that the combined Carry and Excess Carry are very close to the Carry from the Full Running Spread Example.

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# CDS Indices

October 2025

**Dmytro Shelukhin** <sup>AC</sup>

Credit Index & Derivative Strategy

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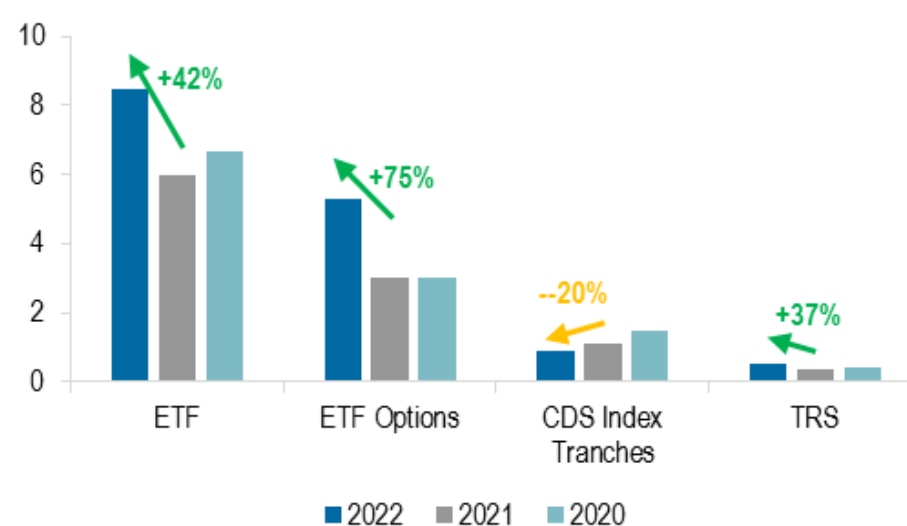
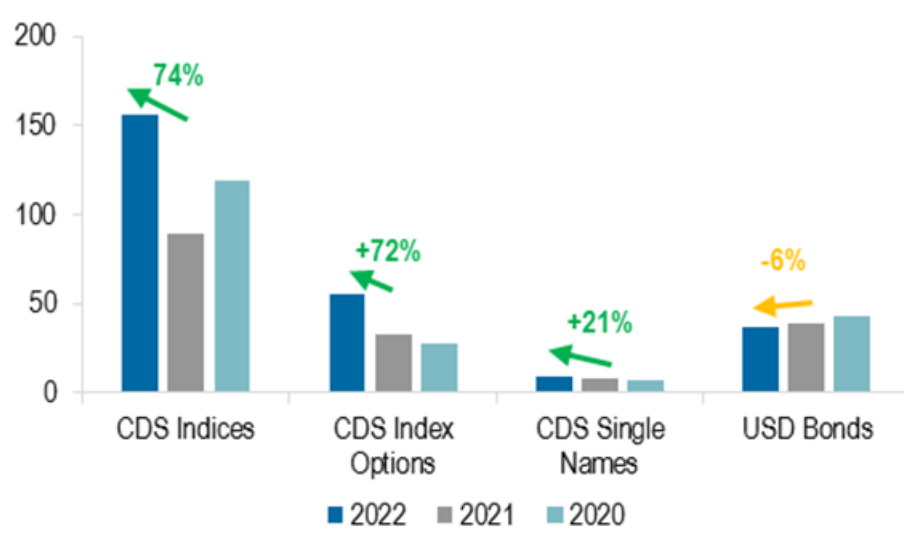
## What is a CDS Index?

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- CDS indices allow investors to take diversified long or short exposure to a credit market.
- CDS indices are the most liquid instruments in the credit markets.
- CDS indices are effectively **baskets of single name CDS**.
  - e.g. iTraxx Main is a basket of 125 single name CDS contracts in European Investment Grade.
- CDS indices have a fixed composition and fixed maturity. New series with an updated basket of underlying credits are launched twice a year (in March and September). Note: old series continue trading.
- CDS indices **trade as a single contract**, meaning that you can get diversified macro exposure to credit without trading a large number of single name CDS contracts.
- (Most) CDS indices are usually **equally weighted**: e.g. each name in iTraxx Main makes up  $1/125^{\text{th}}$  or 0.8% of the index notional.
- The cost of buying protection on iTraxx Main will be very similar to the cost of buying protection on all 125 single names, but there can be some differences (aka index basis or “skew”).
- Why trade CDS indices instead of just the single names?
  - Much easier to trade one single contract than 125 different single name contracts.
  - Lower bid-offer for index.
  - Higher volumes.
  - Quick way to gain exposure to the credit risk of a region or sector.

## CDS Indices are the most liquid instruments in the credit market

### Average Daily Volumes in 2022 (\$bn)



Source: J.P. Morgan, DTCC

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## Global CDS Indices

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### European CDS Indices: iTraxx Europe Family

iTraxx Main (Investment Grade)  
+ Senior / Sub Financials sub-indices  
+ Main ESG  
iTraxx Crossover (High Yield)

### Asia-Pacific CDS Indices:

iTraxx Japan  
iTraxx Asia Ex-Japan IG  
iTraxx Australia

### US CDS Indices: CDX Family

CDX.NA.IG (+ BBB sub-index)  
CDX.NA.HY (+ BB, B, Ex-BB sub-indices)  
CDX Financials (upcoming)

### Emerging Markets CDS

CDX.EM (+ IG & HY sub-indices)

#### Legacy indices:

- iTraxx SovX Western Europe
- iTraxx SovX CEEMEA
- iTraxx Corp CEEMEA
- iTraxx HIVOL
- iTraxx Asia Ex-Japan HY
- CDX Latin America

## Bloomberg screen for CDS Indices: {CDX <GO>}

| 97) Settings ▾              |   |         |        |        |       |      | CDS Index Monitor |       |        |            |      |        |
|-----------------------------|---|---------|--------|--------|-------|------|-------------------|-------|--------|------------|------|--------|
| Markit Indices              |   |         |        |        |       |      | Historical Data   |       | Spread | 3 Months ▾ |      |        |
|                             |   | Spread  | Chg    | Spread | Basis | Roll | Low               | Range | High   | Avg        | +/-  | 3M Chg |
| 1) Americas                 |   |         |        |        |       |      | ◆ Avg ● Now       |       |        |            |      |        |
| 10) CDX Investment Grade    | ⌵ | 47.26   | +0.31  |        |       | 6.0  | 47.2              | —◆—   | 56.3   | 50.8       | -3.5 | -9.1   |
| 11) CDX High Yield          | ⌵ | 107.90* | +0.00  |        | 1.6   | 0.4  | 106.3             | —◆—   | 108.0  | 107.4      | +0.5 | +1.6   |
| 12) MCDX                    | ⌵ | 78.01   | +1.43  |        | 0.0   |      | 78.0              | —◆—   | 84.0   | 82.6       | -4.6 | -3.9   |
| 2) EMEA                     |   |         |        |        |       |      |                   |       |        |            |      |        |
| 20) iTraxx Europe           | ⌵ | 56.20   | +5.88  |        |       | 10.7 | 50.2              | —◆—   | 62.0   | 54.1       | +2.1 | -5.8   |
| 21) iTraxx Crossover        | ⌵ | 268.41  | +20.00 |        |       | 32.9 | 248.6             | —◆—   | 316.0  | 273.4      | -4.9 | -47.6  |
| 22) iTraxx Sr Financial     | ⌵ | 60.12   | +6.55  |        |       | 13.3 | 53.3              | —◆—   | 66.5   | 57.7       | +2.5 | -6.3   |
| 23) iTraxx Sub Financial    | ⌵ | 103.14  | +12.58 |        |       | 23.4 | 90.7              | —◆—   | 117.3  | 99.8       | +3.4 | -14.2  |
| 3) Asia                     |   |         |        |        |       |      |                   |       |        |            |      |        |
| 30) iTraxx Japan            | ⌵ | 56.13   | +5.55  |        |       | 13.3 | 51.5              | —◆—   | 66.9   | 58.1       | -2.0 | -9.7   |
| 31) iTraxx Asia Ex Japan IG | ⌵ | 62.94   | +3.13  |        |       | 11.3 | 60.3              | —◆—   | 81.0   | 69.7       | -6.8 | -18.1  |
| 32) iTraxx Australia        | ⌵ | 60.50   | -1.16  |        |       | 6.6  | 60.5              | —◆—   | 80.5   | 69.3       | -8.8 | -19.4  |
| Emerging Markets            |   |         |        |        |       |      |                   |       |        |            |      |        |
| 40) CDX Emerging Market     | ⌵ | 97.73*  | -0.54  |        |       | -1.3 | 97.0              | —◆—   | 98.4   | 97.8       | -0.1 | +0.8   |
| 41) CDX Emerging Market IG  | ⌵ | 101.68* | -0.01  |        | 0.0   | -0.1 | 101.0             | —◆—   | 101.7  | 101.4      | +0.3 | +0.6   |
| 42) CDX Emerging Market HY  | ⌵ | 94.59*  | -0.01  |        | 1.2   | -0.9 | 93.0              | —◆—   | 95.0   | 94.2       | +0.3 | +1.5   |

Source: Bloomberg

# European CDS Indices

## Inclusion Criteria

### iTraxx Main

- 125 European **Investment Grade** Companies
- Equally weighted
- Composition is based on the names with the highest trading volumes.
- Sector constraints (sector buckets filled based on constituent liquidity within min-max sector budgets):
  - 24-36 Autos & Industrials
  - 20-30 Consumers
  - 16-24 Energy
  - 16-24 TMT
  - 30 Financials
- Constituents must be rated higher than BBB-/Baa3 Neg Outlook – note a stricter definition vs cash indices.
- 1y, 3y, 5y, 7y & 10y points are traded (5y most liquid).
- Quoted on spread.
- Trades on upfront + **100bp** fixed coupon.
- €50bn daily volume across entire market.

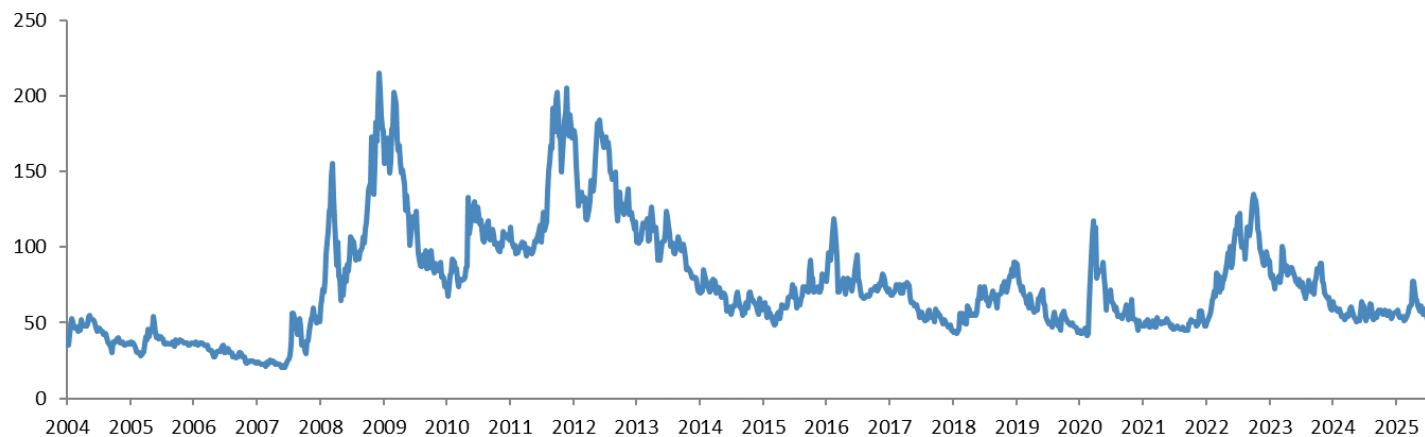
### iTraxx Crossover

- 75 European **High Yield** Companies
- Equally weighted
- Composition is based on the names with the highest trading volumes.
- Financials not allowed (except Specialty Finance)
- No sector constraints
- Constituents must be rated BBB-/Baa3 Neg Outlook or below.
- Unrated issuers are allowed as long as CDS spread is greater than 1.5 x spread of Main Non-Financials.
- No distressed – 500bp upfront < 50pt
- New CDS contracts can be introduced via **Suppl. List**.
- 3y, 5y, 7y & 10y points are traded (5y most liquid).
- Quoted on spread.
- Trades on upfront + **500bp** fixed coupon.
- €15bn daily volume across entire market.

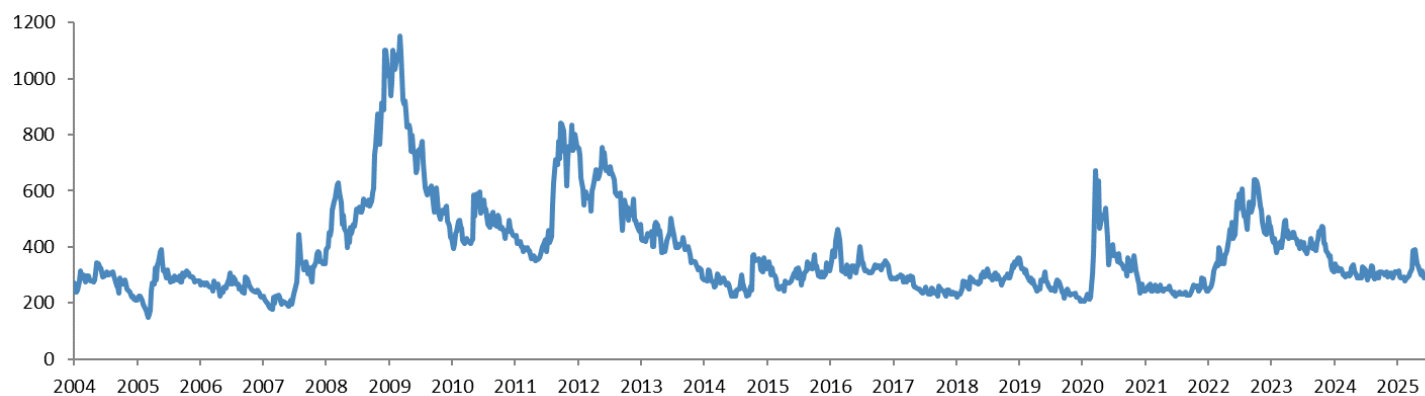
Source: J.P. Morgan

## iTraxx Main and Crossover Historical Spreads

iTraxx Main 5y Spread



iTraxx Crossover 5y Spread



Source: J.P. Morgan



## European CDS Indices - continued

### Inclusion Criteria

#### iTraxx Senior Financials

- 30 European Banks and Insurance Companies
- Equally weighted
- Subset of 30 Financials that appear in iTraxx Main.
- **References the Senior Unsecured debt** (SLAC where available otherwise Senior Preferred/HoldCo).
- 5y and 10y points are traded (5y most liquid).
- Quoted on spread.
- Trades on upfront + 100bp fixed coupon.
- €4.5bn daily volume across entire market.

#### iTraxx Subordinated Financials

- 30 European Banks and Insurance Companies
- Equally weighted
- Same 30 names as Senior Financials.
- **References the Tier 2 debt.**
- 5y and 10y points are traded (5y most liquid).
- Quoted on spread.
- Trades on upfront + 100bp fixed coupon.
- €700m daily volume across entire market.

Source: J.P. Morgan

# North American CDS Indices

## Inclusion Criteria

### CDX Investment Grade

- 125 North American Investment Grade Companies
- Equally weighted
- Very few banks and financials (CDS dealers not allowed)
- Constituents must be rated higher than BBB-/Baa3 Neg Outlook.
- Composition is based on the names with the highest trading volumes.
- 1y, 3y, 5y, 7y and 10y points are traded (5y most liquid).
- Quoted on spread.
- Trades on upfront + 100bp fixed coupon.
- \$60bn daily volume across entire market.

### CDX High Yield

- 100 North American High Yield Companies
- Equally weighted
- Financials not allowed
- Constituents must be rated BBB-/Baa3 Neg Outlook or below
- Composition is based on the names with the highest trading volumes
- Additional mechanism to include new names to increase weightings of under-represented sectors
- 3y, 5y, 7y and 10y points are traded (5y most liquid).
- **Quoted on price**
  - Price = \$100 – Upfront
  - e.g. price of \$99 means buyer pays 1% upfront.
- Trades on upfront + 500bp fixed coupon.
- \$16bn daily volume across entire market.

Source: J.P. Morgan

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## CDS Index Mechanics

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- CDS indices are quoted and traded in the same way as single name CDS.
- *Quoted* on spread basis, which is used to calculate *traded* upfront amount (pv/cash that gets exchanged).
- Example:
  - We buy iTraxx Main protection on €100million for a 5y maturity.
  - Quoted spread is 75bp
  - We will pay the fixed coupon of 100bp.
  - Due to the difference in the quoted spread and fixed coupon we will also receive an upfront amount, equal to 1.125 % of notional.
- CDX.HY is quoted on price, not spread.  $\text{Price} = 100 - \text{Upfront}$ .
- Jargon convention: “buy the index” = buy prot. in iTraxx and CDX.IG but sell prot. in CDX.HY.
- The ISDA Standard Model is used to convert between upfronts and spreads.
- You can download the ISDA Standard Model at [www.cdsmodel.com](http://www.cdsmodel.com) (also implemented in CDSW menu on BBG).

Source: J.P. Morgan

# CDSW calculator on BBG is the industry standard for mark-to-market calculations

XOVER S43 V2 example – with one default (SFR)



Source: Bloomberg

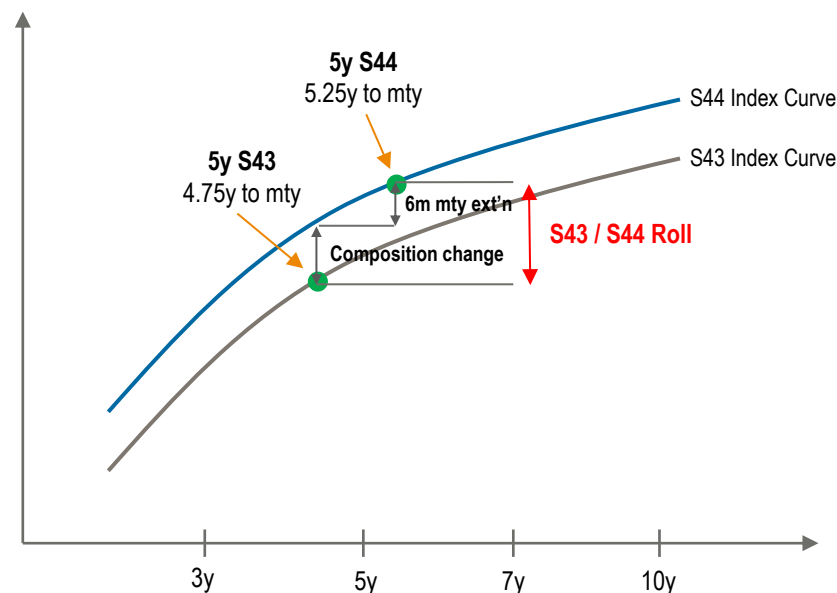
J.P.Morgan

# CDS Index Series and Rolls

- CDS indices trade in what are known as “Series”.
- A new series is launched every six months on **20<sup>th</sup> March** and **20<sup>th</sup> September** – this is known as the “index roll”.
- The new series of each index have a **new composition** and **extra six months of maturity**.
- The latest series is known as the “**on-the-run**” index.
- Currently the “on-the-run” series is Series 44 for iTraxx and 45 for CDX indices.
- For example, on 22 September 2025 we rolled from iTraxx S43 to iTraxx S44.
  - iTraxx S43 indices will have an updated composition reflecting liquidity and rating changes over the previous six months.
  - iTraxx S44 5y maturity is 20-Dec-2030, six months longer than the iTraxx S43 5y maturity of 20-Jun-2030.
- Old indices become “off-the-run”. They still trade but become less liquid over time.

## Rolls in Practice

- Switching a position from an old series to a new series is known as “rolling” the position.
  - “Buying the roll” = rolling buy protection position = “steepener” trade
- E.g., an investor who had bought protection on Main S44 could roll to the new series by selling back protection on Main S43 and buying protection on Main S44.
- Rolls are optional; if investors do nothing then they continue to hold the position in the old series.
- The majority of positions held by investors are rolled to the new series in order to maintain liquidity.



Source: J.P. Morgan

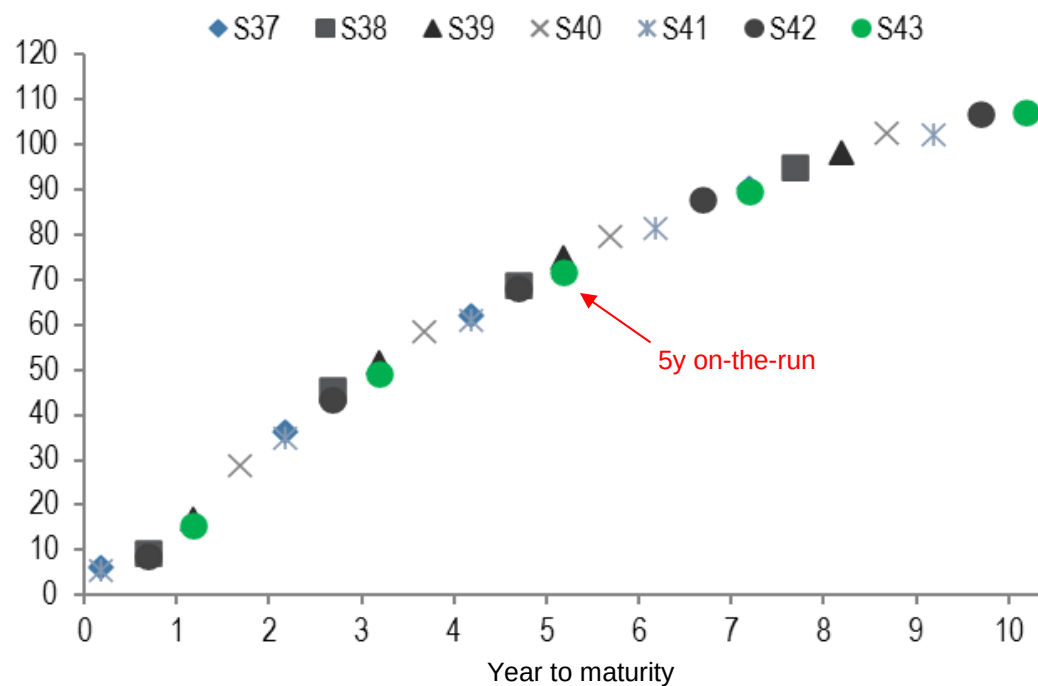
## iTraxx Main S42 / S41 Composition Differences

| ENTRANTS (in S42, but not in S41) |        |         | LEAVERS (in S41, but not in S42) |        |           |
|-----------------------------------|--------|---------|----------------------------------|--------|-----------|
| Name                              | 5y CDS | Reason  | Name                             | 5y CDS | Reason    |
| 1 ALSTOM                          | 96     | Upgrade | GSK PLC                          | 28     | Liquidity |
| 2 Rolls-Royce PLC                 | 79     | Upgrade | DSM B.V.                         | 20     | Liquidity |
| 3 Cellnex Telecom SA              | 81     | Upgrade | United Utilities PLC             | 58     | Liquidity |

Source: J.P. Morgan

## Each index is a “curve”

There isn't just one “iTraxx Main” index contract – although many investors only care about 5y OTR.







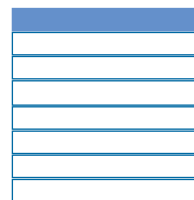
# CDS Indices and Credit Events

## Credit Events

- CDS indices handle credit events in the exact same way that a basket of single name CDS.
- When a credit event happens, the single name splits out from the CDS index and is settled separately like a normal single name CDS contract.
- The defaulted single name is not replaced; the CDS index continues to trade on a reduced notional.
- Coupons will be paid on the full notional until the credit event date, and on the reduced notional afterwards.
- Example:
  - We buy €125mm of iTraxx Main protection (125 names), equivalent to €1mm protection on each of the single names.
  - There is a credit event on one name with a 40% recovery.
  - We receive €600k = €1mm x (1-40%) from the single name portion of the index.
  - The index will continue to trade and will now be known as “Version 2” of the Series. After the auction, the new reduced notional of the index will be €124mm.

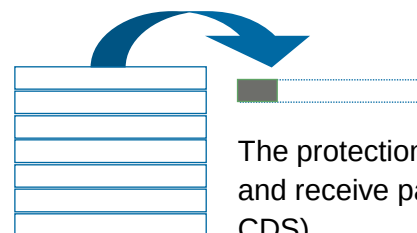
Source: J.P. Morgan

### 1 Credit Event on one of the names



The protection buyer and the protection seller can trigger the contract

### 2 The triggered name is stripped out of the contract and settled separately like a normal single name CDS.



The protection buyer will deliver the bonds and receive par, or cash settle (like a normal CDS)

### 3 The contract remains live on a reduced notional



Premium is paid or received on a reduced notional going forward

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## Traded vs. Theoretical Spread: Basis to Theoretical

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- The traded spread of a CDS index is driven by demand and supply.
- We can also calculate a theoretical spread for the index using the spreads of the underlying single names.
- These spreads will often not be the same – the difference is known as the basis-to-theoretical (aka index skew).
- **Basis to theoretical = Traded Index spread – Theoretical Spread**
- Why does the basis exist?
  - Both macro factors and idiosyncratic factors can be the driving force in market moves.
  - In environments where macro factors are more important, investors are more likely to express views using the index than single names.
    - The index is likely to go wider first during sell-offs (causing the basis-to-theoretical to become more positive) and go tighter first during rallies (basis becomes more negative).
  - In idiosyncratic default risk-led environments single-name hedging drives markets.
    - In this case single name CDS moves wider first during sell-offs (basis becomes less positive / more negative).
- If the basis gets too large, investors will look to arbitrage the difference.

# Trading the Basis

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## Index Skew or Arb lists

- The index and underlying single names hold exactly the same default risk.
- If the basis-to-theoretical is very positive or negative, investors will look to arbitrage the difference by trading both the index and all the underlying single names.
- Example:
  - iTraxx Main is trading at a spread of 100bp and the underlying single names give a theoretical spread of 90bp.
  - Basis to theoretical = +10bp.
  - We sell protection on iTraxx Main at 100bp and buy protection on all 125 single names.
- To make a trade worthwhile the basis needs to be large enough to offset the cost of the bid-offer on the index and single names.
- These trades are known as *basis-to-theoretical trades* or *arb lists*.
- Arb lists in general are difficult to execute, given the problems associated with trying to line up a large number of single name trades simultaneously.

Source: J.P. Morgan

# Calculating the Theoretical Index Spread

## Theoretical Spread

- We can calculate a theoretical index spread using single name trading levels.

- As a starting point, the upfront of the index should be equal to the sum of the upfronts of all the single names, adjusted for their weighting within the index.

$$U_I = \sum_{i=1}^N \omega_i U_i$$

$U_I$  = Index Upfront  
 $U_i$  = Single Name Upfront  
 $\omega_i$  = Weighting of name in index  
 $N$  = Number of names in index

- If the index is equally weighted (as most are), then  $\omega_i = 1/N$ . We can also express the upfronts in terms of risky annuities, spreads and coupons.

$$RA_I (S_I - C) = \frac{1}{N} \sum_{i=1}^N RA_i (S_i - C)$$

$S_I$  = Index Spread  
 $S_i$  = Single Name Spread  
 $RA_I$  = Index Risky Annuity  
 $RA_i$  = Single Name Risky Annuity  
 $C$  = Fixed Coupon

- We can rewrite the index risky annuity  $RA_I$  as the sum of the individual single name risky annuities  $RA_I = \frac{1}{N} \sum_{i=1}^N RA_i$

- This gives us:

$$\frac{S_I}{N} \sum_{i=1}^N RA_i - \frac{C}{N} \sum_{i=1}^N RA_i = \frac{1}{N} \sum_{i=1}^N RA_i S_i - \frac{C}{N} \sum_{i=1}^N RA_i$$

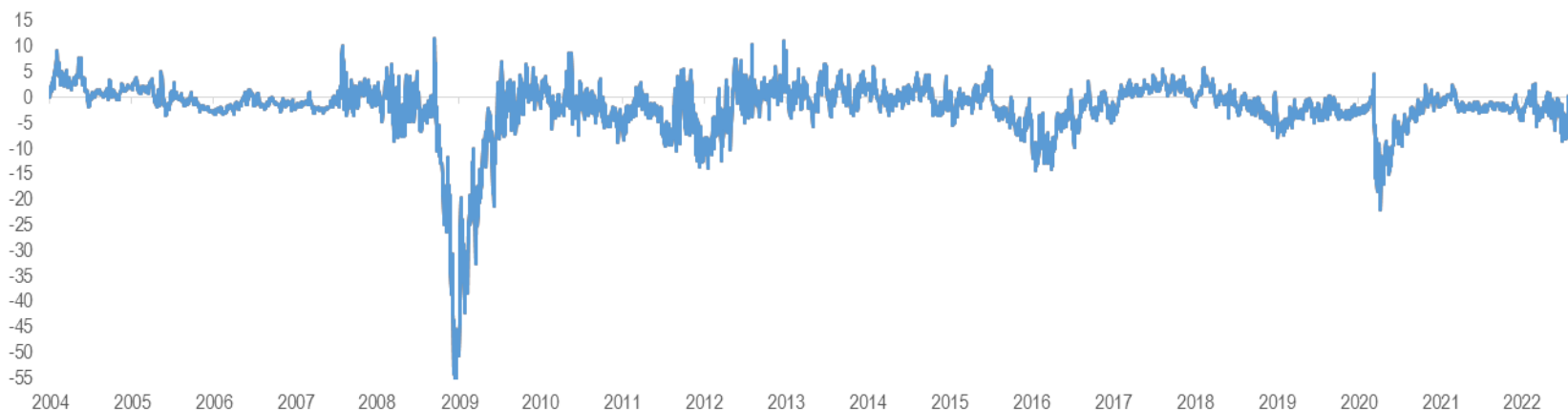
- The coupons cancel out and we can then express  $S_I$  as  $S_I = \frac{\sum_{i=1}^N RA_i S_i}{\sum_{i=1}^N RA_i}$

- **Wider names have lower risky annuities and therefore a lower weighting** – they are more likely to default so we expect to pay the spread for a shorter amount of time.
- The lower weighting of wider spreads means that the Theoretical Spread is lower than a simple average of the single name spreads.

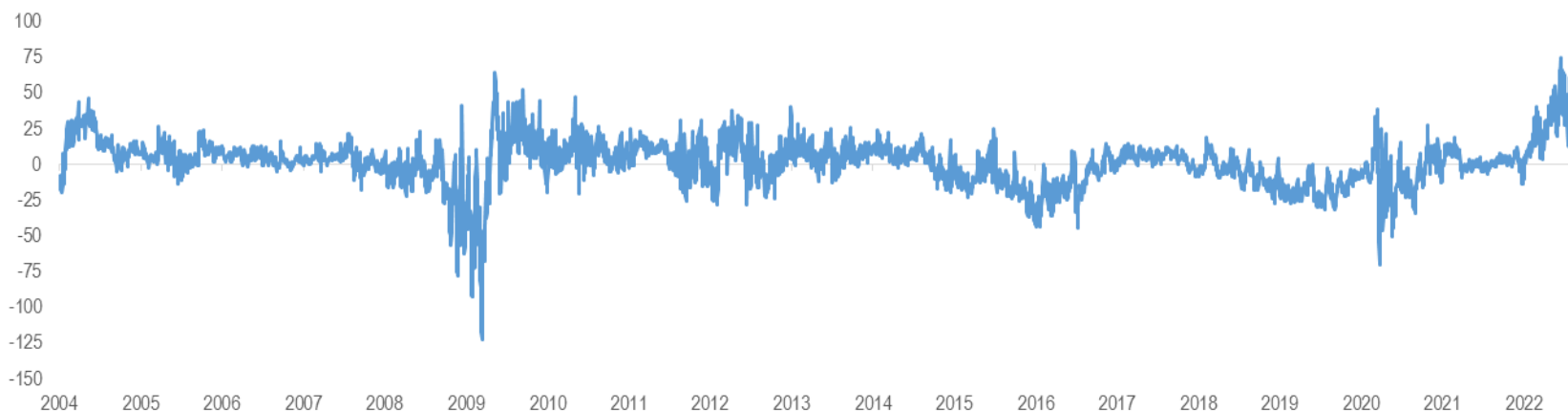
Source: J.P. Morgan

## Historical Basis-to-Theoretical

iTraxx Main Basis-to-Theo (bps)



iTraxx Crossover Basis-to-Theo (bps)



Source: J.P. Morgan

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## Example Relative Value Trades with CDS Indices

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- CDS indices allow investors to take relative value views across different dimensions: time, portfolios, and markets
- The four most popular relative value trades: index basis, curve, roll, and cross-index
- **Index basis trades (skew package):** trading the index vs. the underlying single-name CDS in the portfolio
  - Example: Buy Main Series 40 5y protection and sell 5y protection on the individual 125 single-name CDS in the Series 40 basket
- **Curve trades:** relative value across the tenors of a single index
  - Example: 5s10s CDX.IG on-the-run duration-weighted steepener
- **Roll trades:** relative value across different Series of the same index family
  - Example: Sell protection Xover Series 38 5y and buy protection Xover Series 40 5y
- **Cross-index trades:** relative value between different markets (within credit or cross-asset)
  - Main vs Xover (IG vs HY)
  - Main vs SnrFins (Corps vs Fins)
  - Main vs CDX.IG (Europe vs US)
  - Main vs SX5E (Credit vs Equity)
  - Main vs VSTOXX (Credit vs Equity Vol)
  - Main vs IEAC (Synthetics vs Cash)

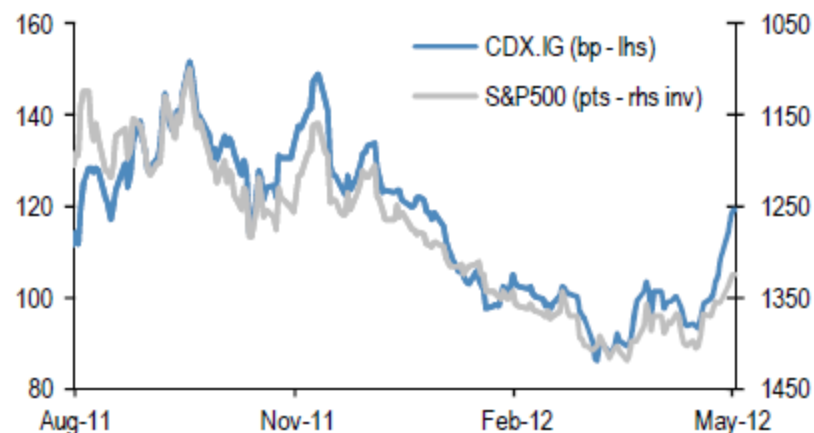


## Example 1 - Cross-Index Trades

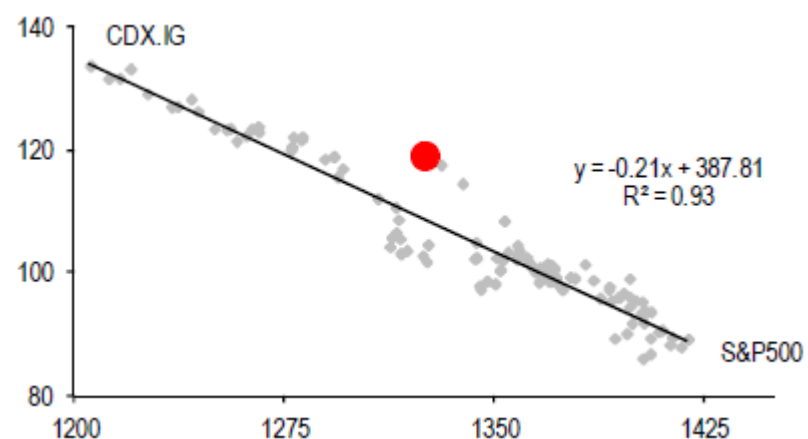
### Credit vs Equity

- The indices are attractive for cross-market trades
  - Credit vs equity: CDX vs S&P 500
  - High Grade vs High Yield: CDX.IG vs CDX.HY
  - Bonds vs loans: CDX.HY vs LCDX
- Example: Sell CDX.IG protection and short S&P 500 in a 6.5:1 ratio
  - In mid 2012, CDX.IG underperformed too much and thus looked too wide vs. S&P 500
  - CDX.IG looked about 7bp too wide vs. S&P 500, 2.5 std dev
  - European risks driving the market but no credit-specific developments, hence both S&P 500 and CDX.IG should trade in line
  - Recommend selling \$100mm CDX.IG protection and shorting 230 S&P E-minis, i.e., long risk CDX.IG vs. short risk S&P 500 in a 6.5:1 ratio

### Mid 2012: Equities outperformed CDX.IG



### ...and CDX.IG seemed wide vs. S&P



Source: J.P. Morgan

J.P.Morgan

## Example 2 - Curve Trades: CDX.IG 5s10s duration-neutral steepener

**CDX.IG looks too flat vs. trend and vs. fair value 5s10s duration-weighted steepener has attractive time value**

### Trade

- Sell CDX.IG 5y protection and buy CDX.IG 10y protection, duration neutral (i.e., sized 1.8:1).

### Rationale

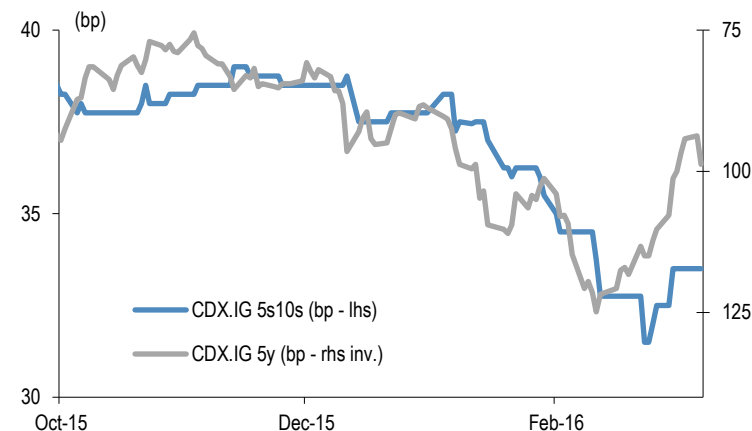
- CDX.IG 5s10s curve appears too flat relative to its recent trend vs. the index and also relative to its underlying single-name CDS. The curve is about 3bp too flat in either perspective.
- By comparison, iTraxx Main is 36bp steep with the 5y index at 92bp.
- The duration-weighted steepener is also carry positive (\$460k per year, if the trade is sized \$180mn:\$100mn) and slide positive (about \$400k per year, assuming the curve is unchanged from its current levels). So the total time value looks quite attractive at about \$860k per year.
- The main risk to the trade is that the markets sell off and the curve flattens as a consequence. The current beta is for the 5s10s curve to flatten by about 1.3bp when the 5y index widens by 10bp. However, the trade would still break even if the curve steepens by about 5bp over the next year as the carry is quite positive.
- There is also a risk in case of near-term default as the trade net sells \$80mn of protection. However, we believe this is unlikely, even though some names in the index are trading quite wide.

### Trade Details

| B/S       | Instrument | Expiry   | Notional    | Entry Level | 1y carry   | 1y slide |
|-----------|------------|----------|-------------|-------------|------------|----------|
| Sell Prot | CDX.IG 5y  | Dec 2020 | 180,000,000 | 99bp        | 1,782,000  | -450,769 |
| Buy Prot  | CDX.IG 10y | Dec 2025 | 100,000,000 | 132bp       | -1,320,000 | 859,054  |
| Net       |            |          |             | 33bp        | 462,000    | 408,284  |

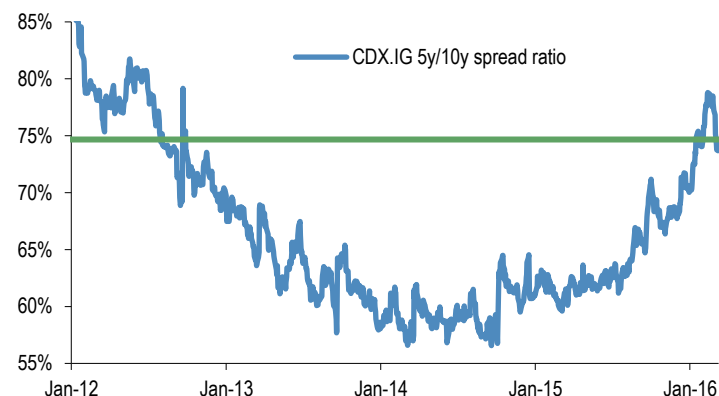
Source: J.P. Morgan. Pricing as of EOD 8 Mar 16.

**CDX.IG 5s10s curve has not steepened by much in the recent rally, and the curve looks too flat vs. the index**



Source: J.P. Morgan

**In relative terms, the 5y level has not been this close to the 10y level in almost four years**



Source: J.P. Morgan

**J.P.Morgan**

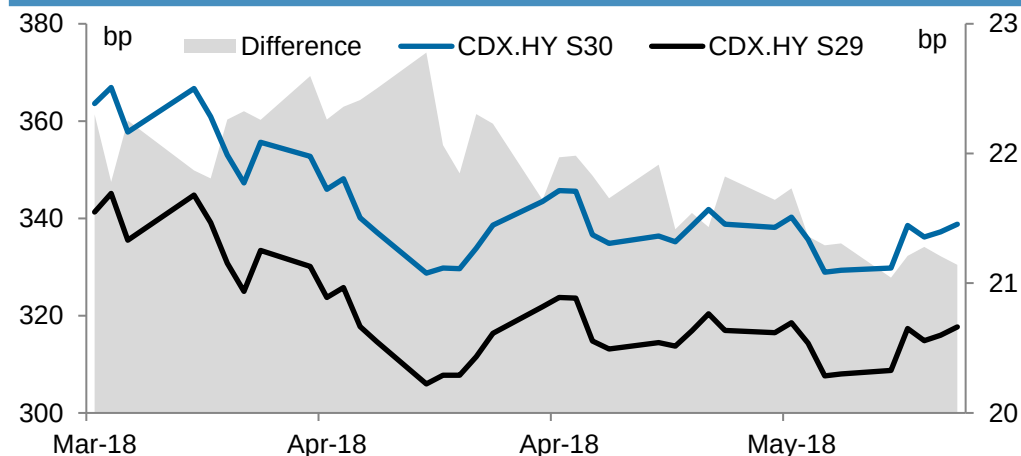
## Example 3 - Roll Trades

- As new CDX index Series are launched every 6 months, off-the-run Series still trade (even though liquidity tends to be larger in the more recent Series)
- The differences between Series are
  - Composition of the portfolio
  - Maturity date
- Investors who have a view between the portfolios can trade one Series vs. another
- Two ways to implement the trade
  - Same maturity date
    - Example: CDX.HY S30 3y vs. S26 5y, which both mature in Jun 2021
  - Same tenor
    - Example: CDX.HY S30 5y vs. S29 5y
    - Note that this is also a curve trade

### Difference between CDX.HY S29 and S30

| Series 29          | 5Y CDS (bp) | Series 30             | 5Y CDS (bp) |
|--------------------|-------------|-----------------------|-------------|
| MBIA Insurance     | 763         | MBIA Inc              | 642         |
| Level 3 Comm       | 188         | Sabre Holdings Corp   | 257         |
| CalAtlantic Group, | 127         | United Continental    | 241         |
| Realogy Group      | 306         | The ServiceMaster     | 337         |
| M.D.C. Holdings    | 154         | Continental Resources | 88          |
| <b>Average</b>     | <b>308</b>  | <b>Average</b>        | <b>313</b>  |

### CDX.HY S30 has traded wider to S29 since inception



Source: J.P. Morgan, as of May 2018

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## Appendix: Additional Material

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- [Credit Derivatives Handbook](#), 2006, J.P. Morgan
- [Global CDS Indices](#), 2025, J.P. Morgan
- [CDS Indices Primer](#), 2023, S&P Global
- [iTraxx Europe and iTraxx Crossover Index Rules](#), 2024, S&P Global
- [CDX.IG and CDX.HY Index Rules](#), 2022, S&P Global

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